



universität
wien

MASTERARBEIT / MASTER'S THESIS

Titel der Masterarbeit / Title of the Master's Thesis

“Bidirectional Mass Spectrometry Modeling Aided by Auxiliary Intermediate Steps From Graph Transformation Tools”

verfasst von / submitted by

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angestrebter akademischer Grad / in partial fulfillment of the requirements for the degree of

Master of Science (MSc)

Wien, 2026 / Vienna 2026

Studienkennzahl lt. Studienblatt / UA 066 910
degree programme code as it appears on
the student record sheet:

Studienrichtung lt. Studienblatt / Computational Science
degree programme as it appears on
the student record sheet:

Betreut von / Supervisor: ao. Univ.-Prof. Mag. Dr. Christoph Flamm

Acknowledgements

I would like to thank my supervisor Christoph Flamm for the opportunity and his patience. For tolerating my occasional slow grasp of concepts and for all the support I received, I am grateful.

Gratitude is also extended to the entire TBI working group for their support and for creating a stimulating environment. Especial thanks to Manuel Uhlir for getting me started and for the many helpful discussions and suggestions along the way.

Heartfelt thanks are due to my mother and grandmother, whose constant faith in me has carried me through many challenges. I also wish to acknowledge my grandfather for the memorable time I was able to share with him.

Finally, thank you, Aila, for your warmth and kindness at every step of this journey.

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Abstract

Mass spectrometry is a cornerstone technique in analytical chemistry, metabolomics, pharmaceutical research and many other areas. The ability to predict a molecular structure from a given spectrum, and vice versa, remains a central challenge. While forward prediction of spectra from known structures is increasingly accurate due to machine learning methods, the inverse problem — predicting possible molecular structures from a given spectrum — remains difficult, ambiguous, and poorly interpretable.

The work proposes the integration of aiding intermediate fragments, which are generated by graph transformation tools, as a strategy to enhance machine learning-based bidirectional translation between molecules and spectra. The aim is to enhance model interpretability and accuracy.

The work will offer a systematic method for integrating graph-based intermediates into machine learning pipelines, potentially improving interpretability and performance of structure $\leftrightarrow$ spectrum translation models.

Zusammenfassung

Massenspektrometrie ist eine grundlegende Technik. Sie wird in der Analytischen Chemie, der Metabolomik, der pharmazeutischen Forschung und in vielen weiteren Bereichen eingesetzt. Die Vorhersage einer Molekülstruktur aus einem gegebenen Spektrum — und umgekehrt — bleibt eine zentrale Herausforderung. Während die Vorhersage von Spektren aus bekannten Strukturen dank Methoden des maschinellen Lernens zunehmend präzise gelingt, ist das inverse Problem — also die Vorhersage möglicher Molekülstrukturen aus einem gegebenen Spektrum — weiterhin schwierig, mehrdeutig und nur schwer interpretierbar.

Die Arbeit schlägt vor, unterstützende Zwischenfragmente, die durch graphbasierte Transformationstools erzeugt werden, als Strategie zu integrieren, um das maschinelle Lernen für die bidirektionale Übersetzung zwischen Molekülen und Spektren zu verbessern. Ziel ist es, die Interpretierbarkeit und Genauigkeit der Modelle zu steigern.

Die Arbeit wird eine systematische Methode zur Integration graphbasierter Zwischenstufen in Machine-Learning-Pipelines bieten, um potenziell die Interpretierbarkeit und Leistungsfähigkeit von Struktur $\leftrightarrow$ Spektrum-Übersetzungsmodellen zu verbessern.

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Chapter 1

Introduction

1.1 Motivation

Mass spectrometry (MS) is one of the most widely used analytical techniques for molecular characterization across chemistry, metabolomics, environmental sciences, and pharmaceutical research. [1, p. 1] Its high sensitivity [2, p. 1764], reproducibility [2, p. 1764], and scalability [2, p. 1764] have enabled large-scale molecular profiling in both research and industrial settings. Modern instrumentation such as Orbitrap, Q-TOF, and triple quadrupole mass spectrometers generates high-throughput spectral data that can be compared against large repositories of molecular fingerprints such as the NIST Mass Spectral Library to support compound identification and profiling [3, pp. 7274-7275].

Despite these advances in data acquisition, the interpretation of MS data remains a significant bottleneck [4, p. 1]. Structural elucidation from spectra is still largely expert-driven, relying on chemical intuition and established fragmentation rules [4, p. 11]. While thousands of spectra can be recorded within a short time frame [4, p. 1], the corresponding structural annotation process remains comparatively slow, manual, and difficult to scale [4, p. 3]. This imbalance creates a fundamental asymmetry between data generation and data analysis.

The intrinsic complexity of molecular fragmentation contributes substantially to this challenge [1, p. 240]. Fragmentation processes are governed by chemically valid bond cleavages and rearrangements [1, pp. 30-31], leading to multiple competing fragmentation pathways. A single molecule may therefore generate numerous fragments through direct cleavages and rearrangement reactions [1, p. 240-241]. Because fragmentation pathways depend sensitively on molecular structure and internal energy [1, p. 26, pp. 30-31], structurally distinct molecules may produce partially overlapping spectral signatures, while small structural variations can induce significant

spectral differences. Consequently, the inverse problem — inferring molecular structure from spectral data — is inherently ill-posed [1, p. 240].

Recent machine learning approaches aim to address this challenge by learning direct mappings between molecular representations and spectral outputs. Graph neural networks [5], sequence-based encoders [6, p. 1573], and generative models have demonstrated promising performance in forward (molecule-to-spectrum) and inverse (spectrum-to-molecule) prediction tasks. However, purely base-data-driven approaches exhibit several limitations:

- They often lack mechanistic interpretability. [7, p. 3& 6]
- They do not explicitly model fragmentation pathways. [7, p. 3]
- They rely on statistical correlations rather than chemically structured priors. [8, p. 966]
- Their generalization to rare or out-of-distribution molecular scaffolds remains limited. [9, p. 12]

In particular, models that treat molecules as flat graphs or string encodings and spectra as unordered peak lists ignore the structured chemical processes underlying fragmentation. This omission reduces interpretability and may hinder robust extrapolation beyond the training distribution. [10, p. 3420]

Fragmentation in mass spectrometry is not an arbitrary stochastic process; it follows chemically constrained transformation sequences. [11, ch. 4& 8] Therefore, incorporating structured representations of fragmentation—such as derivation graphs encoding valid transformation rules—offers a principled way to introduce chemical inductive bias into machine learning models. By explicitly modeling fragment lineage and transformation pathways, it becomes possible to align molecular substructures with spectral evidence in a mechanistically meaningful manner.

This thesis is motivated by the hypothesis that integrating chemically structured derivation graphs into a machine learning framework can improve predictive robustness, interpretability, and bidirectional structure-spectrum modeling. By combining fragment-aware representations with data-driven learning, the aim is to bridge the gap between high-throughput spectral acquisition and scalable, chemically grounded interpretation.

1.2 Objectives

The objective of this thesis is to design, implement, and systematically evaluate a fragment-augmented machine learning framework for improving the bidirectional translation between molecular structures and mass spectra. Specifically, the work investigates how the integration of

graph-based reaction and fragment derivation graphs, generated through chemically valid graph transformation rules, can serve as structured and interpretable intermediate representations that enhance both predictive performance and model transparency.

Mass spectrometry-based structure elucidation represents an inherently ill-posed inverse problem: structurally distinct molecules may yield partially overlapping spectral signatures, while minor structural modifications can induce substantial spectral changes. This thesis therefore introduces fragment-level intermediates derived from rule-based graph transformations to inject chemically grounded structure into the learning process.

Instead of directly learning a mapping between molecular graphs and spectral representations, the proposed framework decomposes the task into a structured pipeline in which molecular graphs are transformed into chemically plausible fragment sets, which in turn mediate the relationship to spectral outputs. These fragment sets are embedded into a shared latent space together with molecular and spectral representations, enabling aligned embeddings across the domains.

Ultimately, this thesis aims to demonstrate that incorporating chemically valid graph-based fragment derivations into neural learning pipelines, enhancing both the predictive power and the interpretability of bidirectional structure-spectrum modeling in mass spectrometry and enabling de novo molecular generation from spectral data.

1.3 Scope and Limitations

This thesis is restricted to electron ionization (EI) mass spectrometry, as EI fragmentation reactions are comparatively well understood and mechanistically characterized. [11, ch. 4& 8] This allows the use of chemically informed rule-based graph transformations to generate fragment derivation graphs under consistent assumptions. Other ionization techniques are excluded due to their differing fragmentation behavior and the additional modeling complexity they would introduce.

A major limitation arises from the computational cost of graph-based fragment generation. Rule-based graph transformations lead to combinatorial growth in possible fragments, [12, p. 4] especially when multiple reaction rules and recursive fragmentation steps are considered. Ensuring chemical validity and avoiding duplicate graph states further increases computational demands. [13, pp. 2-3]

Due to this scalability constraint, the experimental evaluation is limited to order of magnitude of 100 compounds. While sufficient for demonstrating methodological feasibility and analyzing interpretability, this restricted dataset limits chemical diversity and does not allow

for generalization. Consequently, the results should be interpreted as proof-of-concept validation rather than evidence of large-scale applicability.

It is also important to note that the reactions are defined in terms of graph symbols in a ambiguous manner. Furthermore, the transformation rules are not exhaustive of all possible fragmentation pathways or even incorrect. This means that the generated fragment derivation graphs may not capture all relevant chemical transformations, and the model’s performance may be sensitive to the specific choice of transformation rules. Therefore, the interpretability analysis is constrained by the assumptions embedded in the rule set. Consequently, it may not generalize to fragmentation processes that deviate from these assumptions.

Chapter 2

Background and Related Work

2.1 Fundamentals and Capabilities of Mass Spectrometry

Mass spectrometry is a core analytical technique that characterizes chemical species by converting them into gas-phase ions and measuring their mass-to-charge ratio (m/z). [14, p. 5] A mass spectrometer therefore acts as a highly sensitive “molecular balance” whose output – a mass spectrum – reports which ionic species are present and with what relative abundances.

Conceptually, an MS experiment comprises three main steps:

- (i) **ionization**, in which neutral molecules are transferred into the gas phase and charged; [14, p. 1]
- (ii) **mass analysis**, in which ions are separated according to m/z ; and
- (iii) **detection**, in which ion signals are recorded and assembled into a spectrum. [14, p. 5]

Because many compounds form multiple ion species (e.g., adducts [14, p. 21], isotopologues [1, p. 67], fragments), the measured spectrum is not a direct image of the neutral molecule but a structured measurement that must be interpreted in the context of the ionization and fragmentation processes used. [14, p. 16-18]

A key capability of MS is the *identification of unknown compounds*. Even without prior knowledge of the analyte, MS provides constraints that narrow the space of possible identities. Accurate m/z measurements can restrict candidate molecular formulas; isotope patterns (e.g., from ^{13}C , ^{37}Cl , ^{81}Br) further refine these candidates by providing diagnostic abundance ratios and characteristic spacings. [1, p. 263] When fragmentation data are available – either through tandem MS (MS/MS) or through ionization methods that induce fragmentation – structural hypotheses can be tested by matching observed fragment ions and neutral losses to expected

substructures. [1, p. 225] In practice, unknown identification often combines these orthogonal cues with computational searches in spectral libraries and structure databases, where a measured spectrum is matched against reference spectra or predicted fragmentation patterns. [15, p. 8] For complex samples, chromatographic coupling (GC-MS or LC-MS) adds retention time as an additional dimension, [1, p. 475] improving selectivity and reducing confounding effects from co-eluting compounds. [14, p. 233-237]

Also in *purity control and quality assurance* MS serves a critical role, particularly in pharmaceutical manufacturing. By detecting the presence and abundance of impurities, degradation products, and unexpected byproducts, MS ensures that manufactured compounds meet specification and quality standards. The high selectivity and sensitivity of MS make it ideal for identifying trace contaminants that would escape less discriminating analytical techniques. Combined with chromatographic separation, MS provides definitive identification and quantification of both the target compound and potential impurities in a single analytical run. [16, p. 2]

MS is equally important for the *quantification of known substances*, where the goal is to estimate analyte concentration rather than molecular identity. [14, p. 260] Quantitative MS is grounded in the relationship between the number of detected ions and the amount of analyte introduced into the instrument. [14, p. 261 & 265-266] In idealized settings, ion signal intensity scales proportionally with concentration; in real measurements, this proportionality is skewed. Consequently, reliable quantification typically relies on calibration strategies. *External calibration* constructs a response curve from standards of known concentration, while *internal standardization* – often using isotopically labeled analogs – compensates for variability during sample preparation, injection, and ionization. [14, p. 264-268] The outcome is a method that can measure trace concentrations across wide dynamic ranges, which is central in many applications.

Beyond identity and amount, MS provides a systematic route to *deduce structural features of molecules*. Structural inference is primarily enabled by fragmentation behavior: when a precursor ion breaks apart, the resulting fragment ions reflect chemically plausible bond cleavages and rearrangements, thereby encoding information about functional groups, substructures, and connectivity. [17, p. i49] Certain fragmentation pathways are characteristic – for example, losses of small neutrals such as H_2O , NH_3 , or CO can indicate the presence of hydroxyl, amino, or carbonyl functionalities, while specific fragment series can suggest alkyl chain lengths or aromatic substitution patterns. [1, p. 504] Importantly, structural elucidation from MS data is rarely straight forward; rather, it is an inference problem in which multiple candidate structures may explain the same spectral evidence, especially for isomers. [18, p. 53]

As a result, modern workflows increasingly integrate MS measurements with computational models that generate and rank candidate structures based on spectral consistency.

2.2 Principles of Mass Spectrometry

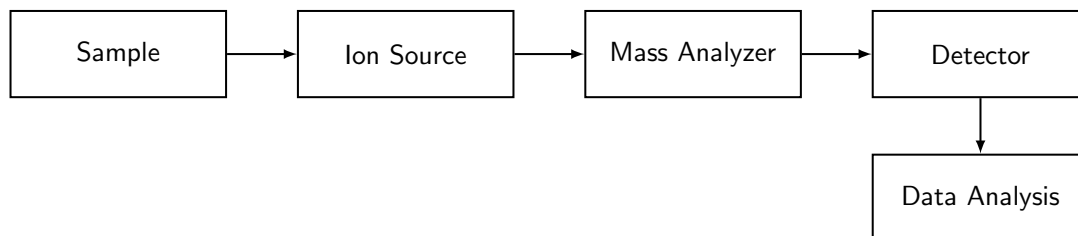


Figure 2.1: Simplified block diagram of a mass spectrometer. The sample is introduced into the ion source, where analyte molecules are ionized. The resulting ions are separated in the mass analyzer according to their mass-to-charge ratio (m/z), detected by the detector, and subsequently processed in the data analysis step.

[19]

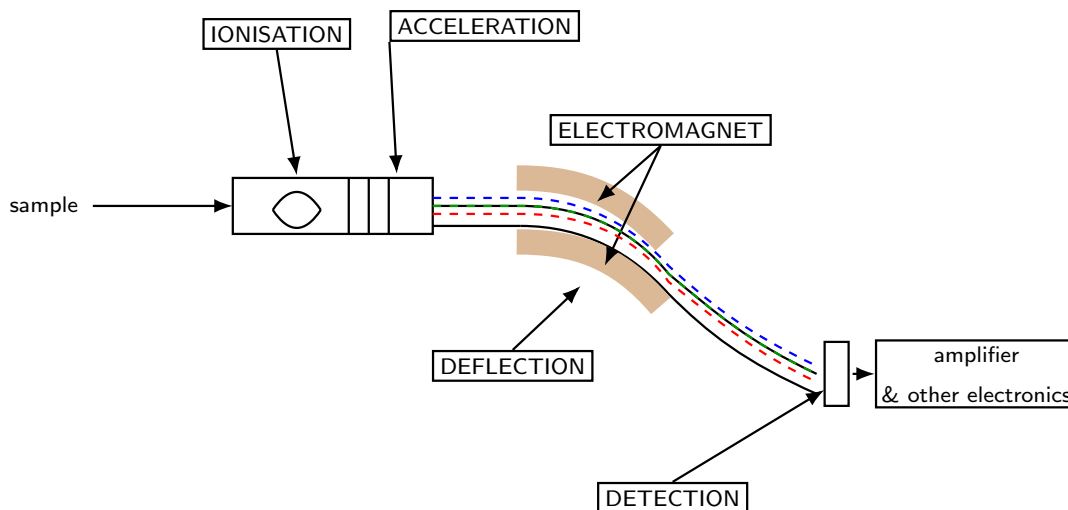


Figure 2.2: Simplified schematic of the measurement principle in a magnetic-sector mass spectrometer. Following sample introduction, ionisation and acceleration generate an ion beam that is separated in the magnetic field by m/z -dependent deflection. The separated ions are then detected and transformed into an electrical signal for further processing.

[20]

Instrument workflow

Figure 2.1 shows steps in a mass spectrometer: a *sample* is introduced, converted into gas-phase ions in the ion source, separated according to their mass-dependent motion in the *mass analyzer*,

detected in the *detector*, and finally interpreted during *data analysis*. The central observable is the mass-to-charge ratio m/z . A mass spectrum is therefore a distribution of signal intensity as a function of m/z , where the signal reflects the abundance of ions reaching the detector. [21, p. 1566] In most practical contexts, intensities are reported as relative intensity, normalized such that the most intense peak equals 100% of the base peak. [21, p. 1550] The x-axis is conventionally labeled with m/z (mass-to-charge ratio), a dimensionless quantity. Labeling with mass units such as *daltons* (Da), amu, or unified atomic mass unit (u) is discouraged – although common in practice – as it causes confusion with multiple charged ions. [21, p. 1516]

Ion notation and precursor concepts

Ions observed in mass spectrometry are typically derived from a neutrally charged molecule M . In electron ionization (EI) a hard ionization method, removal of an electron yields the molecular ion [21, p. 1541] – also called the parent ion – denoted $[M]^+$, [21, p. 1582] a radical cation carrying both a positive charge and an unpaired electron. More generally, the term *precursor ion* refers to the ion selected for fragmentation, [21, p. 1578] from which product ions are generated. In soft ionization methods like electrospray, ion formation is often dominated by proton transfer, producing a protonated molecule $[M + H]^+$. [21, p. 1580]

Peak types: isotope, fragment, and base peaks

A measured spectrum contains several conceptually distinct peak types. *Isotope peaks* arise from naturally occurring heavier isotopes (e.g., ^{13}C , ^{37}Cl), producing characteristic $M + 1$, $M + 2$, ... patterns adjacent to the monoisotopic ion peak. *Fragment peaks* arise when precursor ions undergo dissociation, yielding smaller product ions whose m/z values and relative intensities encode structural information. The *base peak* is the most intense peak in the spectrum [21, p. 1524] and is used for normalization; [21, p. 1550] it is not necessarily the molecular ion and often corresponds to a particularly stable fragment. [22, p. 2] [1, p. 263]

2.2.1 Ionization and Fragmentation

Ionization techniques are broadly classified into two categories based on the energy imparted to the analyte: *hard ionization* methods, which deliver high internal energy and promote extensive fragmentation, [21, p. 1547] and *soft ionization* methods, which minimize fragmentation and favor molecular ion preservation. [21, p. 1591] [23, p. 34]

Hard Ionization methods

Electron Ionization (EI) is a hard ionization technique in which a stream of high-energy electrons (typically 70 eV) collides with neutral molecules in the gas phase. [1, p. 197] The collision transfers energy to the molecule, causing the ejection of an electron and formation of a radical cation $[M^{+\cdot}]$ with both a positive charge and an unpaired electron. [1, p. 194] Because the internal energy of the molecular ion typically exceeds the activation barriers for bond cleavage, the $[M^{+\cdot}]$ ion spontaneously fragments into smaller product ions and neutral molecules. [24, p. 3] As a result, EI mass spectra are characterized by rich fragmentation patterns that reflect the molecular structure. [25]

The deterministic nature of EI fragmentation pathways makes electron ionization particularly valuable for structural elucidation. [26] Characteristic losses of small neutral species provide diagnostic clues about functional groups and connectivity. [27] EI is the dominant ionization method in gas chromatography-mass spectrometry (GC-MS) and remains widely used in small-molecule analysis. [28, p. 1]

Spark Ionization (SI), is a hard ionization technique [29, p. 1], that often utilized to detect trace elements at parts-per-billion (ppb) levels. [30, p. 428] It uses high-voltage up to 100 kV pulsed electric sparks between two electrodes to vaporize, atomize, and ionize conductive or semi-conductive materials. [30, p. 428]

Soft Ionization methods

Electrospray Ionization (ESI) is a prominent soft ionization technique particularly well-suited for coupling with liquid chromatography (LC-MS) and for analysis of large biomolecules. [31] In ESI, a liquid sample is converted into a fine aerosol of charged droplets by applying a high electric potential (typically several kilovolts) between a capillary needle and a counter electrode. [32, p. 3] As the droplets travel toward the mass analyzer, they undergo solvent evaporation and coulombic fission, eventually releasing gas-phase ions into the vacuum of the spectrometer. [32, p. 7] [33, p. 53]

ESI ionization is exceptionally gentle, with minimal or no fragmentation in the ion source. [32, p. 7] This characteristic makes ESI ideal for determining accurate molecular weights of intact molecules. [33, p. 211] Notably, ESI can generate multiply charged ions $[M + nH]^{n+}$ from large molecules, reducing their effective m/z and enabling their analysis on instruments with limited mass range. [33, p. 50]

A curtain gas is often employed to shield the ion path, preventing neutral contaminants and

solvent droplets from reaching the mass analyzer while allowing ions to pass through. [14, p. 42] The sequence from charged droplet formation to gas-phase ion release is shown in Figure 2.3.

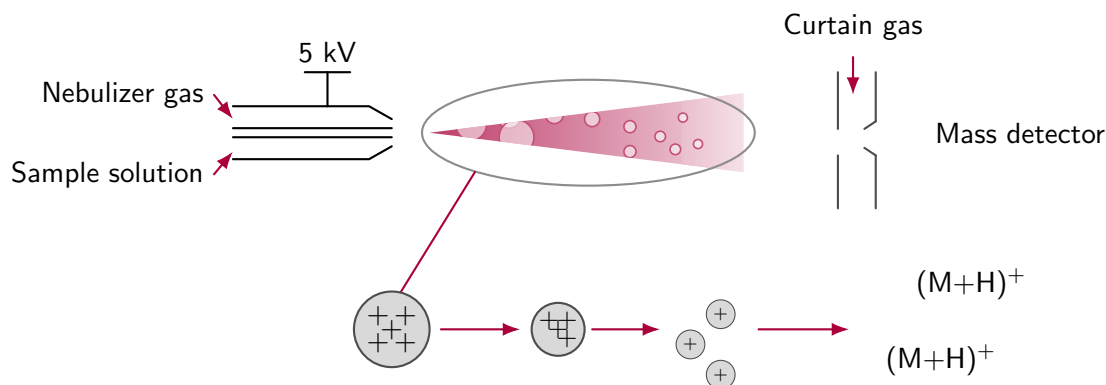


Figure 2.3: Schematic illustration of the electrospray ionization (ESI) process. A sample solution is dispersed by nebulizer gas and subjected to a high electric potential, producing a spray of charged droplets. Progressive solvent evaporation and droplet fission lead to the formation of gas-phase ions, which are transferred through the interface region into the mass detector. The figure highlights the transition from charged droplets to molecular ions, exemplified here by protonated species $(M + H)^+$.

[23, p.39]

Atmospheric Pressure Chemical Ionization (APCI) is a soft ionization method used primarily in LC-MS, [23, p. 39] where analytes are ionized at atmospheric pressure via ion-molecule chemistry initiated by a corona discharge. [21, p. 1522] In contrast to ESI, no high voltage is applied to the inlet capillary; instead, the mobile phase is nebulized, rapidly evaporated in a heated region, [33, p. 56] and the resulting vapor passes a high-voltage needle that generates reagent ions. Solvents dominate the early ionization steps [23, p. 39] and effectively act as a reagent gas, transferring charge to neutral analytes mainly by proton transfer to form $[M + H]^+$.

Because APCI imparts relatively little internal energy, spectra typically show limited fragmentation and are well suited to quantitative LC-MS/MS, [33, p. 171] where signal is concentrated into a few precursor-ion peaks. A heated transfer tube and/or countercurrent curtain gas is used to decluster solvent-analyte adducts prior to ion extraction. [23, p. 39] APCI is often more efficient than ESI for less polar compounds, [1, p. 466] but its higher source temperatures can induce thermal degradation. [23, p. 39]

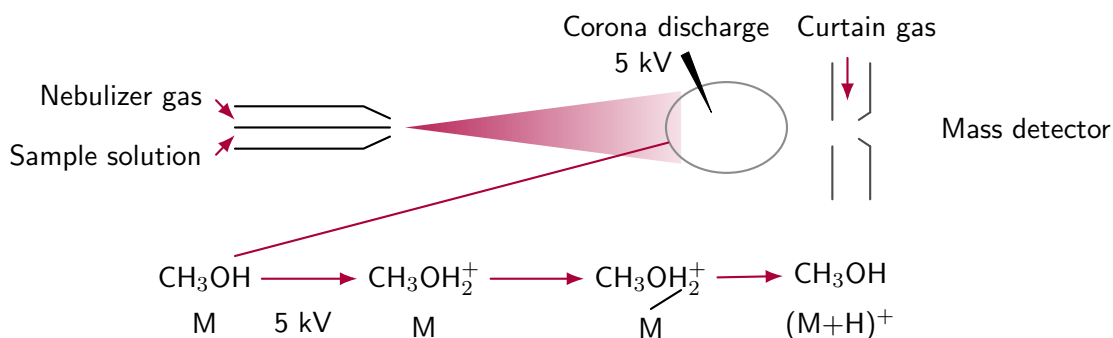


Figure 2.4: Schematic illustration of atmospheric pressure chemical ionization (APCI). The sample solution is nebulized and vaporized, and a corona discharge operated at high voltage generates reagent ions from the solvent. In the example shown, protonated ions transfer a proton to the analyte molecule M , forming the protonated molecular ion $(M + H)^+$, which is then introduced through the interface region into the mass detector.

[23, p.39]

MALDI (Matrix-Assisted Laser Desorption/Ionization) [34] is a soft ionization technique [35]. that uses a laser to desorb and ionize analytes from a solid matrix. [33, p. 36] It is particularly effective for large biomolecules and polymers, [14, p. 33][33, p. 36] producing predominantly singly charged ions. [14, p. 35]. Tuning laser energy and matrix composition can optimize ionization efficiency and control fragmentation. A general schematic of the MALDI process is shown in Figure 2.5.

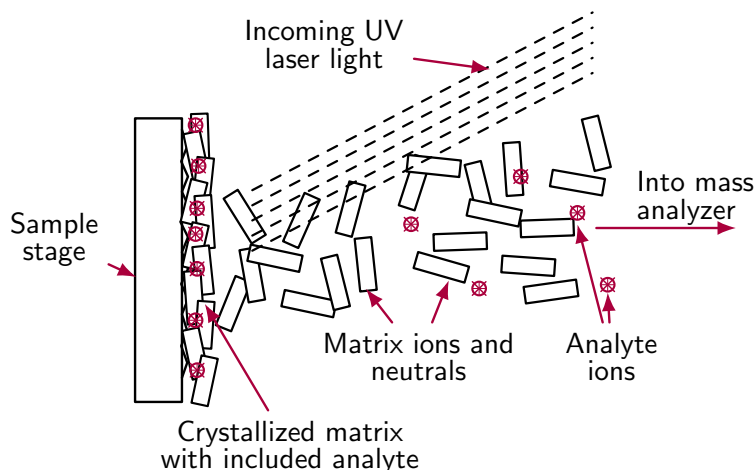


Figure 2.5: Schematic illustration of matrix-assisted laser desorption/ionization (MALDI). A crystallized matrix containing embedded analyte molecules is irradiated with UV laser light at the sample stage, leading to desorption and ionization of matrix and analyte species. During plume formation, matrix ions and neutrals promote the formation of analyte ions, which are subsequently transferred into the mass analyzer.

[23, p. 42]

2.2.2 Mass Analyzers

After ionization, the generated ions must be separated according to their mass-to-charge ratio m/z before detection. This separation is performed by the *mass analyzer*. [33, pp. 5-6] Different analyzer designs exploit electric and magnetic fields to distinguish ions with different m/z values. Among the most widely used analyzers are magnetic sector instruments, time-of-flight (TOF) analyzers, and quadrupole analyzers. [33, pp. 108-109]

Magnetic Sector Mass Analyzers

Magnetic sector analyzers separate ions according to their motion in a magnetic field. The motion of a charged particle with mass m and charge q in electric and magnetic fields is described by the Lorentz force equation

$$\left(\frac{m}{q}\right) \mathbf{a} = \mathbf{E} + \mathbf{v} \times \mathbf{B},$$

where $\mathbf{E}$ is the electric field, $\mathbf{B}$ the magnetic field, $\mathbf{v}$ the ion velocity, and $\mathbf{a}$ the acceleration. Ions with identical mass-to-charge ratios experience identical forces and therefore follow identical trajectories inside the analyzer.

When ions enter a magnetic field perpendicular to their velocity, the Lorentz force bends their trajectory into a circular path. The balance between the magnetic force and the centripetal

force yields

$$qvB = \frac{mv^2}{r},$$

with r being the radius of the circular path, which leads to

$$\frac{m}{q} = \frac{B^2 r^2}{2V},$$

assuming the ions were accelerated through a potential V . Ions with different m/z therefore follow trajectories with different radii and can be spatially separated. [36, pp. 1799-1800]

Time-of-Flight (TOF) Analyzers

Time-of-flight analyzers separate ions according to the time required to travel a fixed distance after acceleration. When ions are accelerated through a potential V , they acquire kinetic energy

$$\frac{1}{2}mv^2 = qV.$$

Solving for the velocity gives

$$v = \sqrt{\frac{2qV}{m}}.$$

In a field-free drift region of length L , the flight time is therefore

$$t = \frac{L}{v} = L\sqrt{\frac{m}{2qV}}.$$

Consequently, lighter ions reach the detector earlier than heavier ions, allowing determination of their m/z values. [37, p. 1152]

Quadrupole Mass Filter (QMF)

Quadrupole analyzers consist of four parallel rods to which a combination of radio-frequency (RF) and direct current (DC) voltages is applied. [38, p. 2] The resulting oscillating electric field produces ion trajectories that are stable only for ions within a specific range of mass-to-charge ratios. Ions with stable trajectories pass through the quadrupole to the detector, while ions with unstable trajectories collide with the rods and are removed. [39, p. 1] By scanning the applied voltages, ions of increasing m/z can be transmitted sequentially, generating a mass spectrum. [38, p. 2]

Quadrupole Ion Trap (QIT)

The quadrupole ion trap operates on the same fundamental principle as the linear quadrupole analyzer described above, added with the ability of oscillating radiofrequency fields to selectively confine ions according to their mass-to-charge ratio. [40, p. 1156] The key distinction lies in the geometry of the electric field: whereas the linear quadrupole uses four parallel rods to create a two-dimensional confining potential that allows ions to pass through along the axial direction, the ion trap employs a three-dimensional quadrupolar field configuration that confines ions in all spatial dimensions within a central cavity. [41, p. 962] This fundamental geometric difference enables the QIT to accumulate and store ions over time [41, p. 973], rather than merely filtering them during transit, which has profound implications for sensitivity, scan modes, and the ability to perform multiple stages of mass analysis within a single device. [41, p. 984]

2.2.3 Detection

After ions have been separated by the mass analyzer, they must be detected and converted into a measurable electrical signal. [33, pp. 5-6] The detector is transforming the arrival of individual ions into quantifiable data that can be assembled into a mass spectrum. [33, p. 103]

The dominant detector in Electron Ionization Mass Spectrometry is the Electron Multiplier (EM). [33, p. 103]

An electron multiplier operates by converting a single incoming ion into a large cascade of electrons through the following sequence:

1. An ion hits the first dynode
2. Secondary electrons are emitted
3. These electrons are accelerated to the next dynode
4. The process repeats (10-20 stages)
5. Result: a measurable current pulse

This cascade amplification principle is illustrated in Figure 2.6.

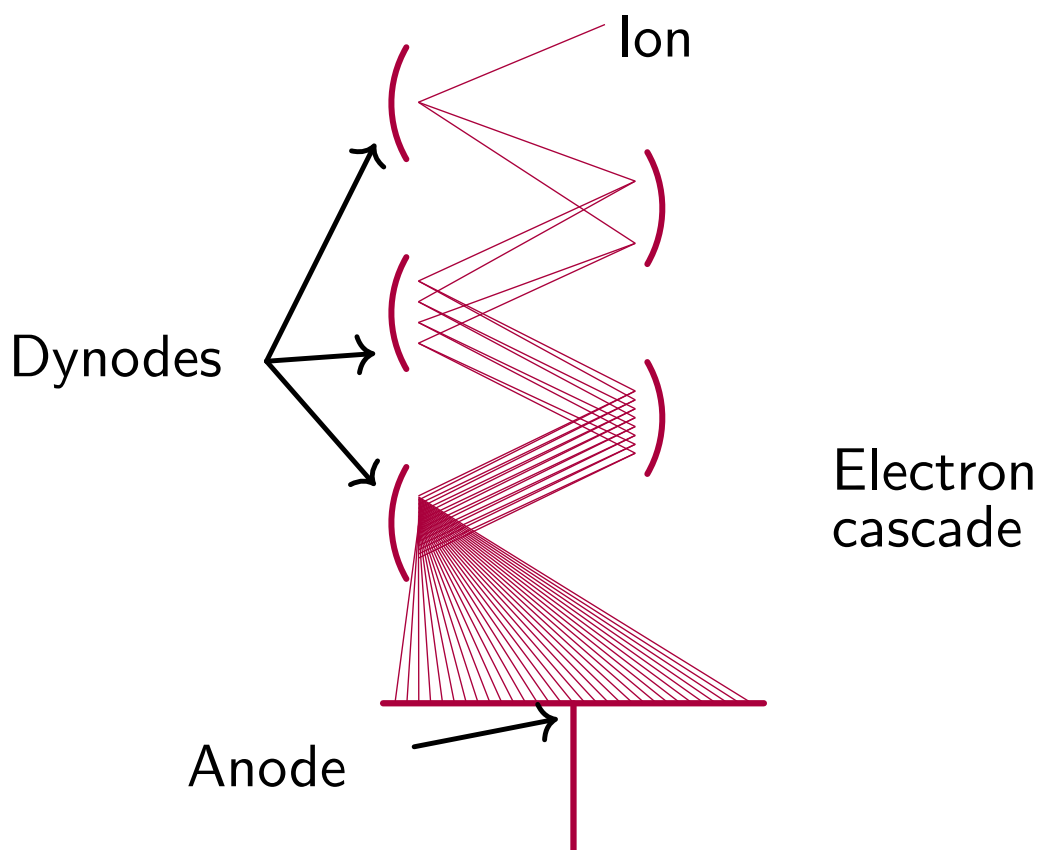


Figure 2.6: Schematic illustration of an electron multiplier (EM) detector. An incoming ion strikes the first dynode, causing the emission of secondary electrons. These electrons are then accelerated to subsequent dynodes, where they generate additional electrons in a cascading effect. After multiple stages, this results in a significant amplification of the original ion signal, producing a measurable current pulse that can be digitized and recorded as part of the mass spectrum.

[23, p. 50]

The amplification factor of an electron multiplier can reach 10^6 to 10^8 , enabling the detection of individual ions with high sensitivity. The resulting current pulses are digitized and recorded as a function of m/z , producing the final mass spectrum that serves as the basis for subsequent data analysis. [33, p. 104]

2.2.4 Manual Interpretation of the MS Spectrum

Data analysis transforms the raw spectrum into structural hypotheses: isotope patterns constrain elemental composition, [1, p. 67] while fragment ions and neutral-loss relationships constrain functional groups and connectivity. [27] Note, multiple candidate structures can be compatible with the same spectral evidence, particularly for isomers under hard ionization methods

like EI. [42, p. 6996]

The following summarize a typical manual workflow [33, pp. 215-216] for interpreting EI mass spectra.

1. Determine the molecular ion peak (M) and its isotopic pattern
2. find isotope patterns (e.g., Cl, Br, S)

Table 2.1: sample isotope patterns

Element	Diagnostic pattern
Cl	3:1 doublet (M, M+2)
Br	1:1 doublet
S	small M+2 peak
C	^{13}C gives predictable M+1

3. determine the base peak (100%) which is the energetically favored fragmentation
4. analyze fragment differences and identify neutral losses
5. match fragmentation to chemical logic defined by reaction rules

These practical heuristics are grounded in specific EI reaction mechanisms, which are outlined in the following subsection.

2.2.5 Reactions in Electron Ionization

Fragmentation reactions in electron ionization mass spectrometry follow well-characterized patterns that are not random but are reproducible pathways determined by [25] the internal energy distribution and the intrinsic chemistry of the molecular ion. [43, p. 846] Understanding these reaction mechanisms is essential for interpreting EI spectra and for predicting fragmentation patterns from molecular structure. [43, p. 846]

They have been studied extensively by McLafferty and Tureček, see [11]. The following subsections describe some of the most important mechanisms.

Homolytic single-bond cleavage

The most straightforward fragmentation pathway is *simple bond cleavage*, in which a single bond in the molecular ion breaks homolytically, yielding a charged fragment and a neutral radical.

The probability of cleavage at a particular bond depends primarily on bond strength: weaker bonds are more readily broken. [1, p. 30] For aliphatic hydrocarbons, C–C bonds adjacent to branching points or heteroatoms are particularly susceptible to cleavage. [1, p. 258]

Alpha Cleavage

Alpha cleavage is a characteristic fragmentation mode in molecules containing functional groups with heteroatoms. [33, pp. 220–221] In this process, the bond directly adjacent to the functional group is cleaved, producing a resonance-stabilized cation. [33, pp. 220–221] For instance, in ketones and aldehydes, cleavage of the C–C bond next to the carbonyl group generates an acylium ion $\text{R}-\overset{+}{\text{C}}=\text{O}$, which is stabilized by the resonance structure $\text{R}-\text{C}\equiv\overset{+}{\text{O}}$. [1, p. 230] This reaction pathway often produces intense peaks in the spectrum and serves as a diagnostic indicator for carbonyl-containing compounds. [1, pp. 234–235]

McLafferty Rearrangement

The *McLafferty rearrangement* is a hydrogen-transfer reaction characteristic of carbonyl compounds and other systems with abstractable γ -hydrogens. [33, p. 223] In the commonly depicted mechanism, a six-membered transition state facilitates the transfer of a hydrogen atom from the γ position to the carbonyl oxygen, accompanied by cleavage of the bond between the α and β carbons. [1, p. 265] The result is an enol radical cation and a neutral alkene. [33, p. 224] This rearrangement is particularly valuable for structural elucidation because it reveals connectivity information that would not be apparent from simple cleavage alone. [33, p. 224]

Charge-Induced Fragmentations

Beyond the radical-site-driven cleavages described above, fragmentation can also be initiated by the charge center itself. In such *charge-directed fragmentations*, the positive charge polarizes nearby bonds, facilitating heterolytic cleavage and producing even-electron cations and neutral radicals. [14, pp. 281]

2.3 Computational Mass Spectrometry

Computational tools significantly enhance MS capabilities for structure elucidation and compositional analysis. [15, pp. 1–2] Modern workflows integrate in silico fragmentation models, spectral database searches, and machine-learning methods to accelerate compound identification and enable high-throughput analysis. [25, p. 1] [15, p. 2]

In this section, we outline the key computational strategies in mass spectrometry, including library matching, in silico prediction, and de novo structure inference, before turning to their mathematical and algorithmic foundations.

2.3.1 Library Search

Library search is a spectrum-matching procedure used to identify an unknown compound by comparing its measured spectrum against a reference database of known spectra. [15, p. 8] Each entry in such a database typically stores a reference spectrum together with metadata such as molecular formula, structure, and acquisition conditions.

The comparison operates on two observable quantities: the m/z values of detected peaks and their relative intensities. A query spectrum is considered a match if a reference spectrum reproduces both the positions and the abundance pattern of its peaks sufficiently well. [23, p. 35]

A typical library search workflow proceeds as follows: [15, pp. 1, 5]

- (i) **Pre-processing:** the query spectrum is converted into a form suitable for comparison, for example by peak selection, intensity scaling, or related preprocessing steps.
- (ii) **Candidate filtering:** database entries may be restricted using precursor mass or precursor m/z information within a tolerance window, reducing the number of candidates that must be evaluated. [5, p. 14]
- (iii) **Scoring and ranking:** each candidate reference spectrum is assigned a similarity score against the query, and candidates are ranked accordingly. [15, p. 5]

The most widely used similarity metric is the *cosine similarity*, computed over the vector of peak intensities aligned by m/z : [44, p. 1]

$$\cos(\mathbf{q}, \mathbf{r}) = \frac{\mathbf{q} \cdot \mathbf{r}}{\|\mathbf{q}\| \|\mathbf{r}\|}, \quad (2.1)$$

where $\mathbf{q}$ and $\mathbf{r}$ are the intensity vectors of the query and reference spectrum, respectively. A score of 1 indicates a perfect match; lower values indicate increasing dissimilarity.

2.3.2 In Silico Fragmentation

In silico fragmentation predicts how a molecule may dissociate in a mass spectrometer and generates a theoretical spectrum from these simulated fragment ions. [7, p. 1] The standard workflow starts from a candidate molecular structure, applies computational fragmentation rules or learned models to produce plausible fragments, converts these fragments into a predicted spectrum, and finally compares the prediction against the measured spectrum using a similarity

score. [7, p. 2] This approach is particularly useful when no exact match is available in spectral libraries, because it enables structure-aware candidate ranking beyond pure reference lookup. [15, pp. 1–2][25, p. 1] At the same time, prediction quality depends strongly on how well the model captures real instrument conditions, ionization-dependent chemistry, and fragment intensities; consequently, performance can degrade for compounds or fragmentation pathways that are underrepresented in the training data or incompletely described by rule sets. [45, p. 1989]

When these *in silico* predictions remain ambiguous, the workflow may shift to *de novo* inference directly from spectral constraints.

2.3.3 De Novo Structure Prediction

De novo structure prediction denotes the inference of molecular structure from scratch, i.e., without relying on pre-existing templates or premade reference repositories [46, p. 1]. Importantly, *de novo* characterizes the *template-free* nature of the workflow and does not, by itself, imply a specific modeling paradigm (e.g., physics-based versus data-driven). Accordingly, *de novo* approaches may incorporate empirical heuristics, statistical learning, or mechanistic constraints, depending on the application. [47, pp. 701–702]

2.3.4 First Principles

In contrast, *first-principles* modeling emphasizes deriving predictions without empirical parameterization, i.e., without fitting to experimental data. [48, p. 1] In physics and chemistry, such first-principles treatments are often realized via *ab initio* electronic-structure methods.

Ab Initio MS Prediction

Ab initio MS prediction refers to computational modeling of mass spectra directly from molecular structure using nonempirical quantum-mechanical methods that are, in principle, independent of experimental fitting [49, p. 1921]. In this sense, *ab initio* approaches constitute first-principles modeling, because they aim to predict fragmentation behavior and resulting spectra from the underlying physics and chemistry of the molecule rather than from learned correlations. Operationally, *ab initio* MS prediction typically relies on quantum-chemical calculations (to obtain energetics and feasible fragmentation channels) combined with dynamical or statistical treatments to relate accessible pathways to predicted fragment distributions.

While *ab initio* methods can provide mechanistic insight and may achieve high accuracy for small molecules, their computational cost often limits scalability to large molecular libraries and complex systems. [47, p. 702]

2.4 Computational Representations of Molecules and Reactions

The encoding of molecular structures and reactions in a machine-processable form is essential for the feasibility of making algorithmic predictions. [50, p. 518]

In computational chemistry, molecules can be represented as graphs. [50, p. 521] Reactions are modeled as graph transformations that break and form bonds. [51, p. 18]

2.4.1 Molecular Graphs

To enable algorithmic processing of molecular structures, we represent them using an abstract graph model that separates topology (which atoms are connected) from optional geometric detail. [51, p. 18]

Figure 2.7 illustrates the abstraction of a molecule as a graph, where vertices represent atoms and edges represent chemical bonds.

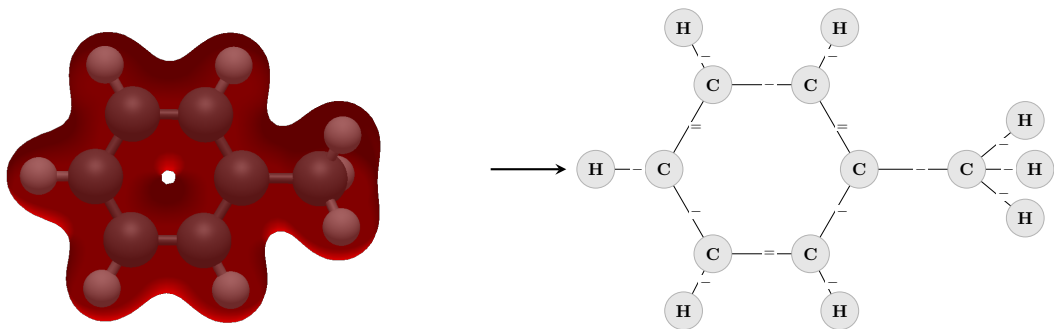


Figure 2.7: Toluene on the left depicted as a quantum-mechanical model with 3D coordinates and electron density, and on the right as a molecular graph abstraction. The graph captures the connectivity of atoms (vertices) and bonds (edges) while abstracting away geometric and electronic details.

This abstraction enables precise operations such as matching and rule-based transformation. [12, pp. 2–3] The formal definition used throughout this thesis is introduced next.

Formal definition

A molecular graph is typically defined as [52, p. 2] [53, p. 3]

$$G = (V, E) \tag{2.2}$$

where:

- V ... Vertices represent atoms

- E ... Edges represent chemical bonds between atoms

In Representations each node may carry attributes such as element type (e.g., C, N, O), formal charge, aromaticity, hybridization state, and isotope label. Likewise, each edge may encode bond order (single, double, or triple), aromatic character, and bond stereochemistry. [54, pp. 4–5] By construction, this basic molecular graph does not include geometric or electronic detail, such as 3D coordinates, molecular conformation, or wavefunction-level information; these require augmented graph models or separate geometric and electronic representations. [54, pp. 3–4] [55, p. 595]

2.4.2 Chemical Graph Transformation Systems

To model chemical reactions algorithmically, molecular graphs are combined with rewrite rules that encode local bond changes. [56, p. 76] In this view, a reaction is not an arbitrary edit of a graph, but a constrained transformation that preserves valid molecular structure while modifying a specific reaction center. [12, p. 2] [57, p. 6]

Double Pushout Graph Rewriting

A standard formalism is the double pushout (DPO) approach. [12, p. 3] A rule is given by

$$p = (L \xleftarrow{l} K \xrightarrow{r} R),$$

where L is the left-hand side pattern to be matched, R is the right-hand side pattern after rewriting, and K is the interface that is preserved. [56, p. 75] Given a match $m : L \rightarrow G$ in a host graph G , DPO rewriting constructs an intermediate graph D (deletion pushout complement) [58, p. 680] and then the product graph H (insertion pushout). Intuitively, parts in $L \setminus K$ are removed, parts in $R \setminus K$ are added, and K remains unchanged. [12, p. 3] [59, p. 4]

$$\begin{array}{ccccc} L & \xleftarrow{l} & K & \xrightarrow{r} & R \\ m \downarrow & & \downarrow & & \downarrow m^* \\ G & \xleftarrow{l^*} & D & \xrightarrow{r^*} & H \end{array}$$

Figure 2.8: Schematic of the double pushout (DPO) graph rewriting process. A rule p consists of a left-hand side L , an interface K , and a right-hand side R . Given a match $m : L \rightarrow G$ in the host graph G , the DPO approach constructs an intermediate graph D by removing parts of G corresponding to $L \setminus K$, and then constructs the product graph H by adding parts corresponding to $R \setminus K$. The interface K is preserved throughout the transformation, ensuring that the resulting graph H maintains valid molecular structure.

In chemistry, this provides a representation of a reaction center: atoms and bonds in the preserved interface K define shared context, while changes between L and R encode bond breaking/forming events. [60, p. 5]

Derivation Graph

Building on the local rewrite step formalized above, derivation graphs – induced by repeated rule application – provide a global view that shows mechanistically plausible chemical states. [12, pp. 2–3]

A derivation graph, often also called a reaction network, is a higher-level graph that captures all transformation steps generated by a graph rewriting system. [61, Sec. 10] [57, p. 2] Whereas a molecular graph represents the internal structure of a single molecule, a derivation graph represents the *space of possible reactions* between many molecular states. [12, pp. 2–3]

Formally, each node corresponds to an object state (e.g., a molecule, multiset of molecules, or intermediate), and each directed edge corresponds to an application of a rewrite rule that transforms one state into another. [56, p. 75] If a rule p transforms state G into state H , this step is represented as

$$G \xrightarrow{p} H.$$

By collecting many such derivations, one obtains a directed graph whose topology encodes reachable states, alternative pathways, and cycles.

In computational chemistry, derivation graphs are useful for enumerating reaction pathways, identifying branching points, analyzing network connectivity, and ranking plausible mechanistic routes under additional constraints (e.g., energy, rule priorities, or experimental evidence). They therefore provide a natural bridge between rule-based chemistry models and downstream tasks such as pathway search, and machine-learning-based scoring of candidate transformations. [57, p. 2]

One example derivation graph for the fragmentation rules applied is in Figure 2.9.

2.5 Machine Learning in Mass Spectrometry

2.5.1 Problem Formulations

Two main problem formulations are common in the literature:

- **Structure-to-Spectrum Prediction:** Given a molecular structure, predict the corresponding mass spectrum. This is a forward problem that can be approached using mechanistic models, data-driven models, or hybrid approaches that combine both.

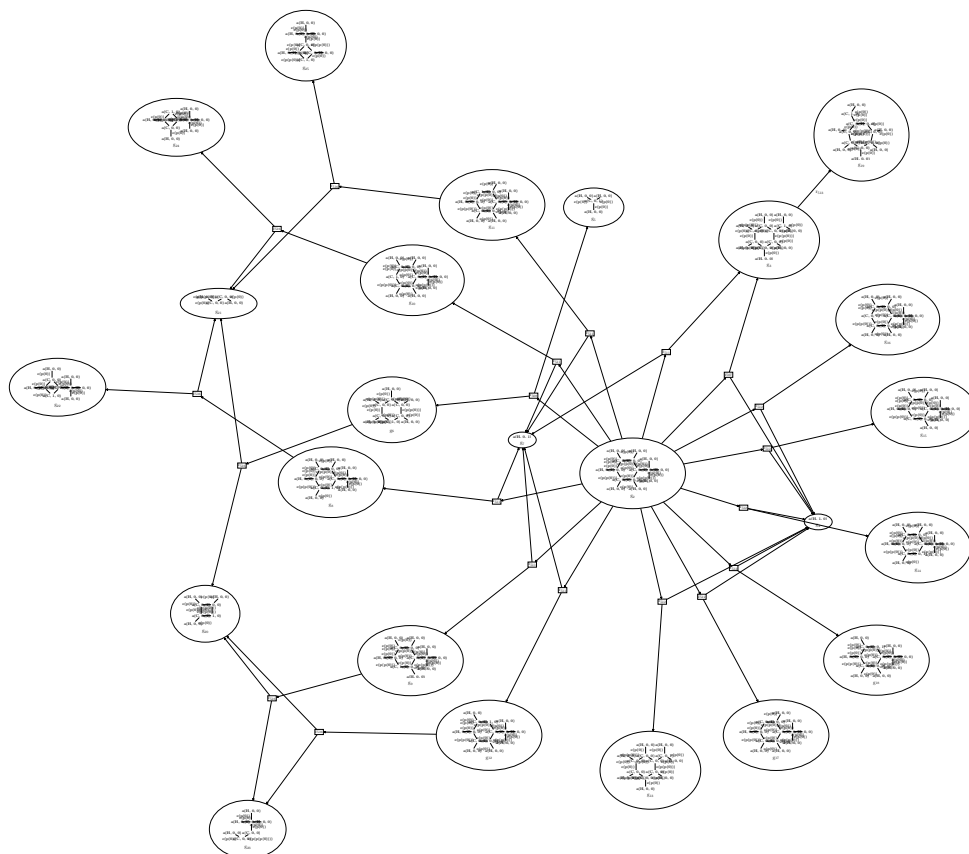


Figure 2.9: Example derivation graph showing reaction pathways for toluene fragmentation. Also illustrates the combinatorial explosion of possible pathways as rules are applied iteratively.

- **Spectrum-to-Structure Prediction:** Given a mass spectrum, predict the corresponding molecular structure. As this is an inverse problem, it is often ill-posed, with multiple candidate structures potentially consistent with the same spectrum.

Both tasks are important for different applications: structure-to-spectrum prediction is useful for *in silico* fragmentation and library generation, while spectrum-to-structure prediction is essential for *de novo* identification of unknown compounds.

Bidirectional models that can perform both tasks are also possible, and may leverage shared representations to improve performance on both tasks.

2.5.2 Case Study: Generating Molecular Fragmentation Graphs with Autoregressive Neural Networks

A prominent example of integrating physical fragmentation models with neural networks is the work of [8], which proposes an approach to predict tandem mass spectra by directly modeling fragmentation pathways as graphs. Rather than treating spectrum prediction as a black-box prediction task, this method learns to generate fragmentation graphs sequentially: at each step,

the model predicts which bonds are most likely to break in the molecular ion, scores the resulting fragments, and iteratively builds a complete fragmentation landscape.

This hybrid approach is particularly relevant to computational MS because it bridges two historically separate paradigms: rule-based *in silico* fragmentation, which is interpretable but computationally expensive and relies on heuristic intensity prediction, and purely data-driven neural networks, which are fast but lack chemical interpretability. [8, p. 3420] By grounding fragmentation prediction in explicit bond-breaking events, the model maintains mechanistic transparency while leveraging the expressiveness of neural networks for learning complex fragmentation preferences from data.

The method demonstrates several key strengths. Because predictions are generated as explicit fragmentation pathways, the model enables human-interpretable explanations: chemists can inspect why particular fragments appear and evaluate whether predicted breakages are chemically plausible. Also the approach shows particular utility for natural products and complex scaffolds where fragmentation patterns deviate from those captured by simple bond-breaking rules, improving metabolite identification from candidate databases. [8, p. 3427]

However, limitations remain. Prediction quality still depends on the distribution of training data; compounds or fragmentation pathways underrepresented in training tend to be mispredicted, and the model may struggle with instrument-specific quirks or unusual ionization regimes not seen during learning.

2.6 Interpretability in Chemical Machine Learning

2.6.1 Why Interpretability Matters

Interpretability in machine learning refers to the ability to understand and explain why a model makes a particular prediction. [62, p. 1]

For structure elucidation from MS data, an uninterpretable model that ranks candidate structures is of limited value if chemists cannot understand the reasoning. Conversely, models that expose their decision-making process [62, p. 2] —by highlighting which structural features drive spectral predictions or which fragmentation pathways are favored— enable hypothesis generation and accelerate scientific insight.

Furthermore, interpretability serves as a diagnostic tool: when model explanations reveal chemically implausible patterns (e.g., incorrect fragmentation rules or unphysical feature importance), this signals potential model failures or training-data biases that numeric performance metrics alone might not expose.

2.6.2 Classic and General Interpretability Methods

Saliency Maps and Gradient-Based Attribution

Saliency maps compute the gradient of the model output with respect to input features, visualizing regions of input space where small changes produce large changes in prediction. For molecular inputs, saliency can be backpropagated through graph neural networks to identify atoms or bonds whose perturbation most affects the predicted spectrum. [63, p. 4] This approach is computationally efficient and provides local explanations for individual predictions. However, gradients can be noisy, especially near decision boundaries, and may not capture higher-order interactions or saturating effects. [64]

SHAP and Permutation-Based Methods

SHAP (SHapley Additive exPlanations) provides a principled, game-theoretic framework for feature attribution based on Shapley values from cooperative game theory. [65, p. 3] By computing the marginal contribution of each feature when iteratively included or excluded, SHAP assigns consistent importance scores that sum to the difference between the model prediction and a baseline. [65, p. 2] For molecular graphs, SHAP can be applied to atomic features or subgraph patterns, revealing which structural motifs drive predictions. Permutation-based importance methods offer similar insights by repeatedly shuffling features and measuring the resulting drop in model performance. [66, p. 23] These methods are model-agnostic and theoretically grounded, but computationally expensive for large molecules or high-dimensional representations.

Counterfactual Explanations

Counterfactual explanations identify minimal modifications to an input that would change the model output, answering the question: “What would need to change for a different prediction?” [67, p. 4] In the chemical domain, counterfactuals correspond to structural modifications—the loss or gain of functional groups, changes in substitution patterns, or alternative regioisomers—that would yield different predicted spectra. [68, pp. 2–3] These explanations are human-interpretable and actionable, but their discovery is computationally challenging, especially in high-dimensional chemical spaces with complex constraints (e.g., synthetic accessibility, valence rules).

2.6.3 Evaluation of Explanations

The quality of explanations must be evaluated in order to determine whether a method actually achieves its explainability objective and to enable comparison between alternative explanation

methods. [69, p. 1] Zhou et al. describe explanation quality in terms of *interpretability* and *fidelity*. Interpretability concerns human understandability and includes the properties of *clarity*, *parsimony*, and *broadness*, whereas fidelity concerns how accurately the explanation reflects model behaviour and includes *completeness* and *soundness*. [69, p. 3]

Based on this view, explanation quality can be assessed through both human-centred and functionality-grounded evaluation. Human-centred evaluation includes application-grounded and human-grounded studies and often considers subjective measures such as trust and confidence together with task-related performance. Functionality-grounded evaluation instead relies on formal proxy metrics tied to specific explainability properties. [69, p. 8] [69, p. 10]

2.6.4 Evaluation Metrics of Prediction Performance

In computational mass spectrometry, evaluation metrics quantify how well predicted spectra match experimental observations. Cosine similarity measures spectral alignment by computing the dot product of normalized peak intensity vectors, accounting for both peak positions and intensities. [70, p. 4] Edit-distance metrics quantify the minimum number of peak modifications needed to transform one spectrum into another, reflecting the structural differences between molecules. [71, pp. 1–2] Wasserstein distances provide a framework for comparing peak distributions, measuring the cost of optimally transporting one spectral peak distribution to another. [72, pp. 1–8] [73] These metrics collectively enable rigorous assessment of model predictions against ground-truth spectra and facilitate comparison of different spectral prediction methods in computational workflows.

Chapter 3

Methodology

3.1 Datasets

This work draws from two main sources of input: measured mass spectra with their corresponding molecular structures as SMILES strings and a set of known fragmentation rules for the ionization method of interest.

3.1.1 Public Databases

For this work, the National Institute of Standards and Technology (NIST) database for EI GC-MS was selected as the primary data source for the measured mass spectra. NIST is considered the gold standard in the field due to several key advantages: [74, p. 3]

- Comprehensive coverage of compounds
- Standardized data format ensuring consistency across entries
- Spectra experimentally measured under controlled, standardized instrument conditions

Alternative databases such as MassBank, PubChem, HMDB, and MoNa were considered but not utilized in this study.

3.1.2 Custom Dataset Construction

3.2 Preprocessing - Generating the Synthetic Data

Synthetic training data are generated with mod derivation graphs (Figure 3.1). In this representation, nodes correspond to molecular states (fragments), while edges represent individual fragmentation transformations. The resulting graph captures the reaction network and preserves

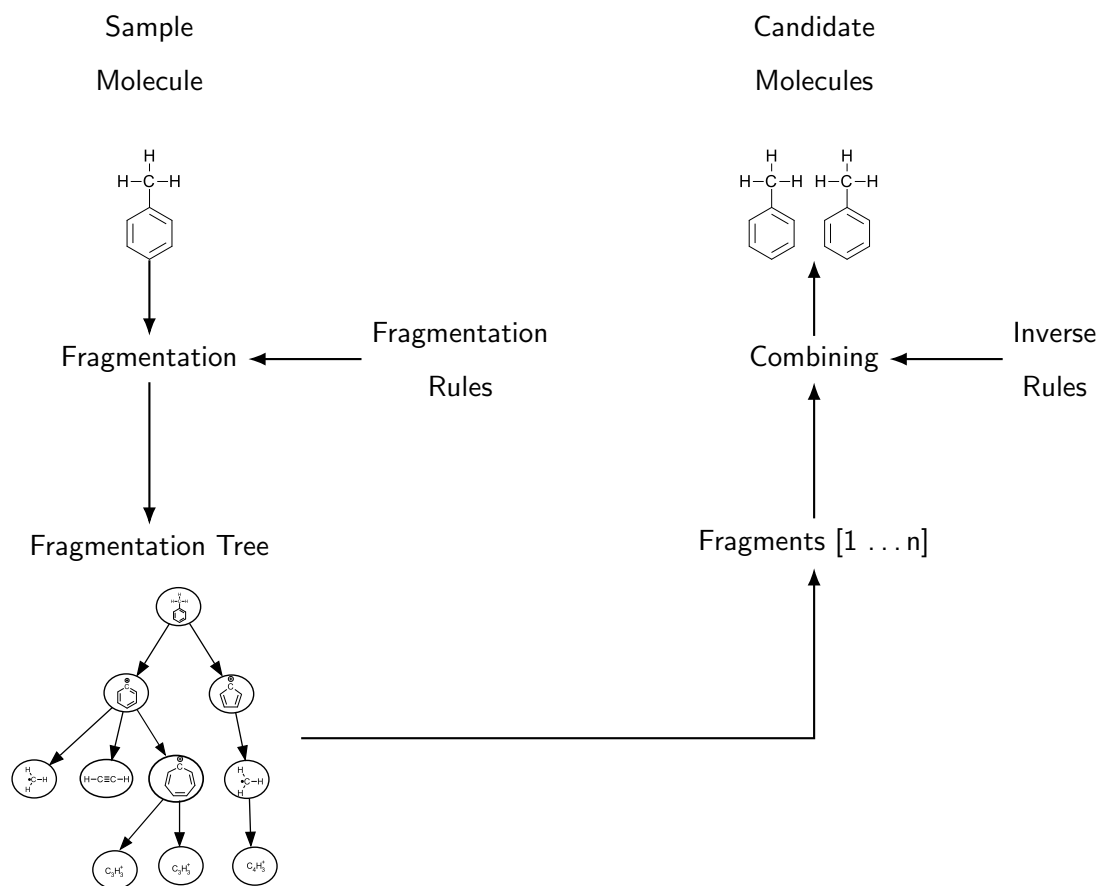


Figure 3.1: Schematic overview of the bidirectional fragmentation workflow. Starting from a sample molecule, fragmentation rules are applied to generate a fragmentation tree that represents possible fragment formation pathways (left). The resulting fragments are then used as inputs for the inverse direction, where candidate molecules are reconstructed by combining fragments according to inverse rules (right). This links forward molecular fragmentation with inverse molecular assembly in a single interpretable framework.

fragment genealogy, which allows us to trace parent-child relationships and identify reachable fragments. This information is later used to validate predicted fragment assignments against chemically plausible pathways.

The **mod** strategy framework provides a systematic approach to explore chemical space by guiding fragmentation pathways through both physico-chemical properties and experimental data, such as mass spectra. A key advantage of this framework is its automatic generation of hypergraphs (reaction networks). This automation enables the construction of detailed atom flow networks for complex fragmentation patterns, making it feasible to track chemical transformations and validate fragmentation predictions at scale. [60, p. 7]

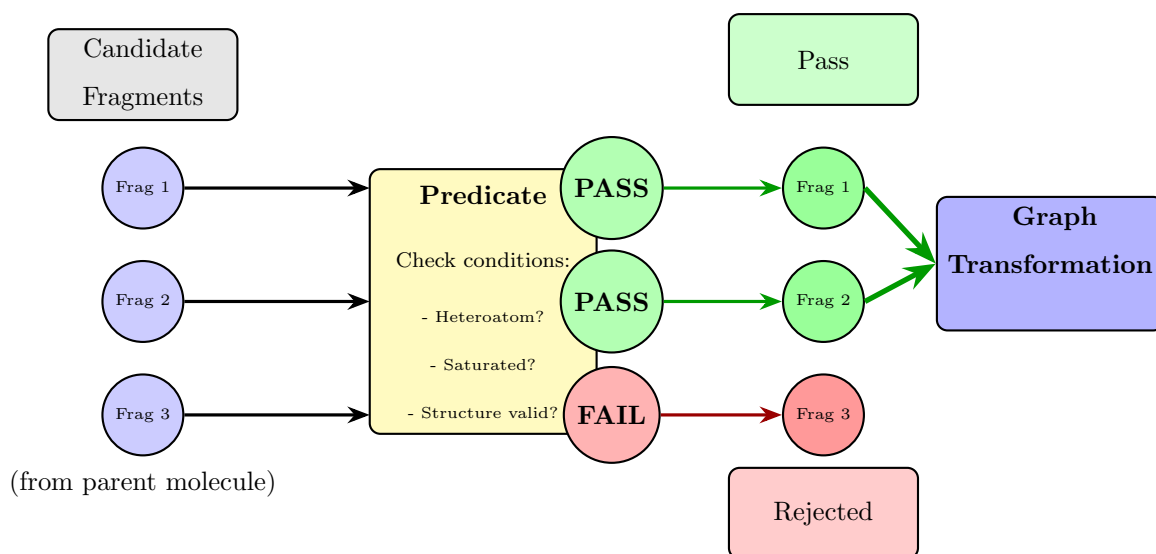


Figure 3.2: Flow diagram explaining predicate filter application at the fragmentation process. During rule application, candidate sites are generated based on the rule’s extension pattern. Each candidate is then evaluated by a series of predicate checks (heteroatom, saturation, alkyl)

3.2.1 Fragmentation Rule Application Strategies

The strategy layer accepts fragmentation rules as input and utilizes them to construct the corresponding reaction network by applying them to molecular structures. Listing 1 shows one representative rule definition used in this workflow.

```

1 IMS_4_9 = mod.Rule.fromDFS(
2     s =
3     "[C]1[C]2[C]3([C]4)[_A+.]5"
4     ">>"
5     "[C.]1" "." "[C]2[C]3([C]4){=}[_A+.]5",
6     name =
7     "radical induced (alpha-)cleavage for a saturated site"
8     " 4.9"
9     " §R1R3R4Y5"
10 )

```

Listing 1: Sample Fragmentation Rule [75, p. 59]

In the `mod` framework, predicates are callback-based filters evaluated during rule application. They remove candidate sites that violate chemical constraints encoded by the rule extension (see Listing 1, line 9). The decision flow of this filtering step is summarized in Figure 3.2.

To operationalize these chemical constraints in the filtering stage, three predicate checks are applied:

Heteroatom Check In order to ensure chemical plausibility during the application of the rule, the heteroatom check retains exclusively those candidate sites which contain at least one non-carbon, non-hydrogen atom within their local neighborhood.

It shall be noted that McLafferty’s original [75, p. 85] rule definition does not explicitly specify the heteroatoms, but one should assume that only the organic elements B, C, N, O, P, S, F, Cl, Br, and I are considered valid heteroatoms in this context. That is not implemented in code, but it is a reasonable assumption given the typical scope of organic mass spectrometry and the nature of fragmentation rules.

The heteroatom check evaluates the neighborhood of the rule-extension location via local graph traversal. A candidate is accepted only if at least one non-carbon, non-hydrogen atom is found, formulated as $\text{BFS} \setminus \{\text{H}, \text{C}\} \neq \emptyset$. If the explored neighborhood contains only carbon and hydrogen, the candidate is rejected. Figure 3.3 contrasts both cases: a valid neighborhood containing a heteroatom and an invalid neighborhood without one.

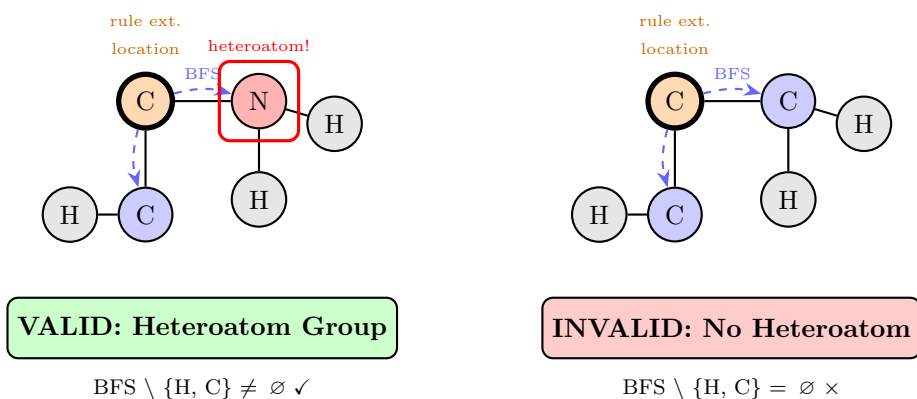


Figure 3.3: Heteroatom check predicate: valid case with detected heteroatom (left) versus invalid C/H-only neighborhood (right).

Saturation Check The saturation check is formulated as a constrained path search between a start and an end atom. A candidate path is accepted only if it follows a carbon backbone through single C–C bonds, while side branches are limited to hydrogen. This removes candidates that include incompatible branch chemistry, such as additional carbon side branches. Figure 3.4 highlights the accepted saturated path in green and marks a rejected carbon branch.

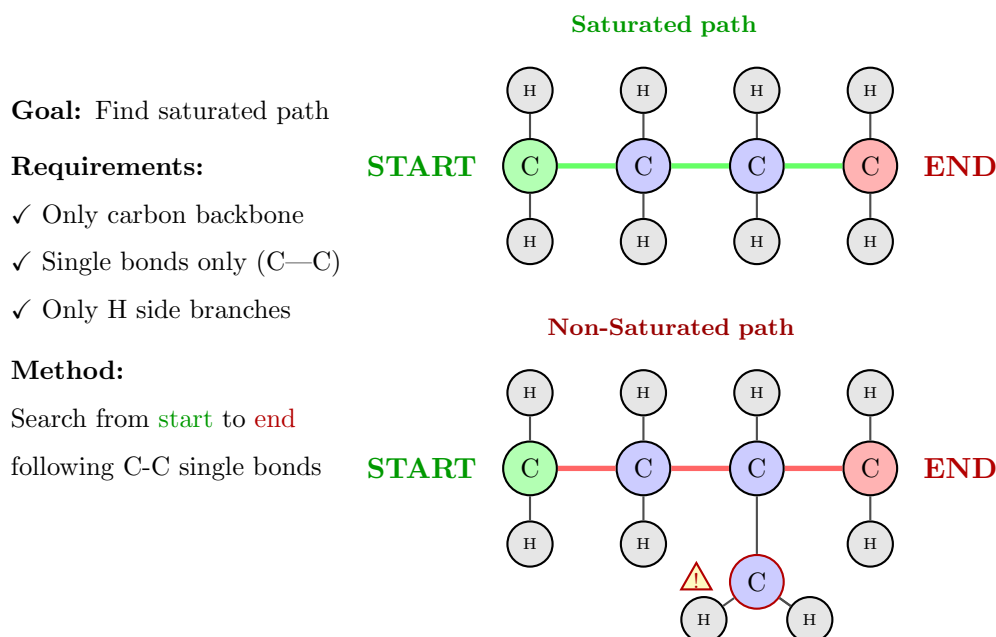


Figure 3.4: Illustration of the concept of a saturated path in a molecular graph. A valid saturated path connects the start and end atoms through a carbon backbone consisting exclusively of single C—C bonds, with only hydrogen atoms as side branches (top). In contrast, a path is not considered saturated if additional carbon side branches are attached to the backbone (bottom).

Alkyl Check The alkyl check ensures that the fragmentation site is part of an alkyl group, restricting fragmentation to aliphatic carbon chains, where such reactions are most likely to occur. This constraint helps filter out fragmentation events in aromatic rings or other structural contexts where different chemical reactivity dominates.

Figure 3.5 visualizes the alkyl check as a graph-constrained traversal: a valid path must connect start and end atoms through only single C—C bonds in a carbon backbone, while side branches are restricted to hydrogen. The green route, which is highlighted, satisfies these constraints; however, the carbon branch, which is marked, is rejected.

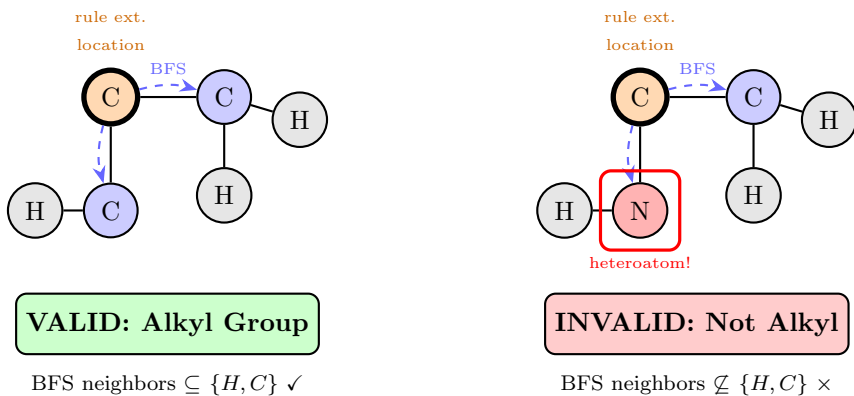


Figure 3.5: Alkyl check predicate formulated as a constrained path traversal. Starting from the rule extension location, a breadth-first search (BFS) is performed over the local molecular graph to verify that all reachable neighboring atoms belong to the set $\{C, H\}$. A matched substructure is accepted as an alkyl group if only carbon and hydrogen atoms are encountered (left), whereas the presence of a heteroatom such as nitrogen causes the predicate to fail (right).

3.3 Latent Representation Schemes

Overview

In this work, we encode molecular fragmentation data through three tightly coupled representation streams:

1. **Molecular states** as chemistry-aware graphs
2. **Derivation dynamics** as a hypergraph over molecular states and transformation rules
3. **Mass spectra** as peak-intensity vectors aligned with structure-derived latent states

The core design principle is a two-level encoding pipeline: first, each molecular state is encoded at atom and bond level; second, these state-level encodings are used as node features in a derivation hypergraph. The model enables the preservation of local chemical structure, while explicitly representing multi-step fragmentation logic.

In particular, a molecule is not encoded as a SMILES sequence, fingerprint, or direct spectrum-oriented tokenization. Instead, it is represented as an explicit graph object derived from the MØD framework, then mapped into a dense latent embedding, and finally placed into a derivation-hypergraph context together with learned rule representations.

3.3.1 Molecular Graph Encoding

PyTorch Geometric (PyG) [76] is a library that extends PyTorch to enable efficient processing of graph-structured data. It provides specialized data structures and operations optimized for graph neural networks (GNNs).

For molecular graph encoding, each molecular state is converted by `GraphFeaturizerMOD` into a PyG `Data` object. Atoms are represented as nodes and bonds as edges, allowing GNN architectures to learn from both node features (atomic properties) and edge features (bond information).

Node features are deliberately minimal and chemistry-focused. For each atom, the feature vector contains:

- atom identity (`atom_id`), the atomic number Z (number of protons)
- formal charge (`charge`), the atom’s formal charge
- radical state (`radical`), whether the atom carries an unpaired electron

Hence, the node representation is

$$x_i = [\text{atom_id}, \text{charge}, \text{radical}].$$

Edge features encode bond type as one scalar value, with single, double, triple, and aromatic bonds mapped to 1.0, 2.0, 3.0, and 1.5, respectively. Connectivity is stored in `edge_index` in COO format.

Before featurization, preprocessing attempts to detect term-graph inputs and converts them into standard graph form where needed. The objective of the normalization step is to ensure the encoder receives a consistently formatted molecular graph representation.

The library handles common challenges in graph processing, such as varying graph sizes and irregular connectivity patterns, through batching mechanisms that stack multiple graphs into a single tensor representation.

For this work, we utilize PyG’s graph convolutional layers to embed molecular structures into a latent space that captures chemical properties while remaining efficient to compute. The flexibility of PyG’s API allows seamless integration with standard PyTorch components for downstream prediction tasks.

3.3.2 Two-Level Molecular Latent Representation

After graph featurization, each molecular `Data` object is passed to a learned graph encoder. In `DGHypergraphFeaturizer`, this is implemented in `_encode_graphs`, where batches of molecular

graphs are processed by `self.graph_encoder`. The output is a dense latent vector of dimension `dim.h` per molecular graph. These embeddings are stored as `mol_rep`.

The central architectural distinction is that molecules are not treated as isolated instances. Each molecular state corresponds to a vertex in a derivation graph (`mod.DG`), and this derivation graph is transformed into a hypergraph representation. The method `_collect_graphs` iterates over derivation-graph vertices, extracts the corresponding molecular graph, and maps each graph to a contiguous vertex index. Each state graph is encoded, producing one latent embedding per derivation-graph vertex. Consequently, node features in the final hypergraph are not raw atom features, but learned embeddings of complete molecular states.

This yields two explicit representational levels:

1. Lower level: each molecule is a molecular graph (atoms and bonds).
2. Higher level: each molecular state becomes a node in a derivation hypergraph.

Technically, molecule encoding proceeds as follows: first chemistry-aware graph featurization, then graph-to-vector compression into a dense latent embedding, and finally use of this embedding as the node feature in the derivation hypergraph.

In the final `HyperGraphData` object generated from the derivation graph, `x` stores molecular embeddings (`mol_rep`), `edge_index` stores incidence pairs (`vertex_id, hyperedge_id`), and `edge_attr` stores learned rule embeddings associated with hyperedges. Thus, `x` can be interpreted as per-vertex graph embeddings, `edge_attr` as per-hyperedge rule embeddings, and `edge_index` as the vertex-hyperedge incidence structure.

3.3.3 Rule Representation and Hyperedge Annotation

Rule processing is implemented separately from molecule processing. Rules are featurized by `RuleFeaturizerMOD`, but unlike molecular states, a rule is not expanded into a chemically rich multi-node graph. Instead, each rule is represented compactly as a one-node `Data` object, whose single node feature is the indexed rule ID.

Within `DGHypergraphFeaturizer`, rule handling is split into three stages. Rules are first collected in `_prepare_rules`. They are then encoded in `_encode_rules`. Finally, they are aggregated in `_pool_rules`. If multiple rules are associated with one hyperedge, their embeddings are mean-pooled into one hyperedge-level vector. As a result, each hyperedge is annotated with a learned representation of the transformation rule (or rule set) that produced that derivation step.

3.3.4 Spectral Vector and Embedding Representation

Each mass spectrum is treated as a set of tuples $(m/z, I)$, where m/z is the mass-to-charge ratio and I is the corresponding peak intensity. For model input, these tuples are converted into peak-intensity vectors over m/z bins. These vectors are embedded into a spectral latent space, which is trained to be compatible with the structure-derived latent representation. The utilization of a shared latent view facilitates the integration of fragmentation topology, molecular-state embeddings, and observed spectral evidence in downstream learning tasks.

3.3.5 Disadvantages

Despite these representational advantages, the proposed encoding pipeline also introduces several methodological and practical limitations:

- Loss of information due to dimensionality reduction from molecular graphs to fixed-size latent vectors
- Limited interpretability of latent dimensions and pooled rule embeddings
- Potential propagation of preprocessing artifacts into learned representations
- Risk of over-squashing and long-range dependency loss in message-passing encoders

3.4 Model Architecture

3.4.1 Overview of the Learning Framework

The implemented learning framework is a bidirectional model that couples molecular structure, fragment-level information, and mass spectra in a shared latent space. As illustrated in Figure 3.6, the model operates on three synchronized inputs: a molecular graph representing the precursor structure, a fragment-derived representation encoding the fragmentation space associated with that molecule, and a binned mass spectrum. These three views are processed by dedicated encoders and are then linked through a shared latent representation that supports both forward spectrum prediction and backward structure retrieval.

On the molecular side, the precursor is represented as a graph and embedded by a graph neural network. On the fragment side, the model uses a derivation-graph backend, in which fragment graphs and their parent-child relations are encoded explicitly. On the spectral side, the measured spectrum is represented as a fixed-length vector obtained by binning intensities over the m/z axis and is projected into the same latent space through a dedicated spectrum encoder.

Therefore, the resulting architecture combines a chemically grounded graph representation with a vector-based spectral representation in a single trainable framework.

A central design decision is that the model is not a direct spectrum-to-SMILES generator. Rather, it is a bidirectional embedding model. The forward direction predicts spectra from molecular and fragment information, while the backward direction maps spectra into a latent space in which candidate molecular structures can be retrieved efficiently. This distinction is important for interpreting Figure 3.6: the right-hand side of the figure should not be read as a decoder that explicitly reconstructs a molecular graph, but as a retrieval-oriented inverse path that aligns spectral embeddings with molecular embeddings.

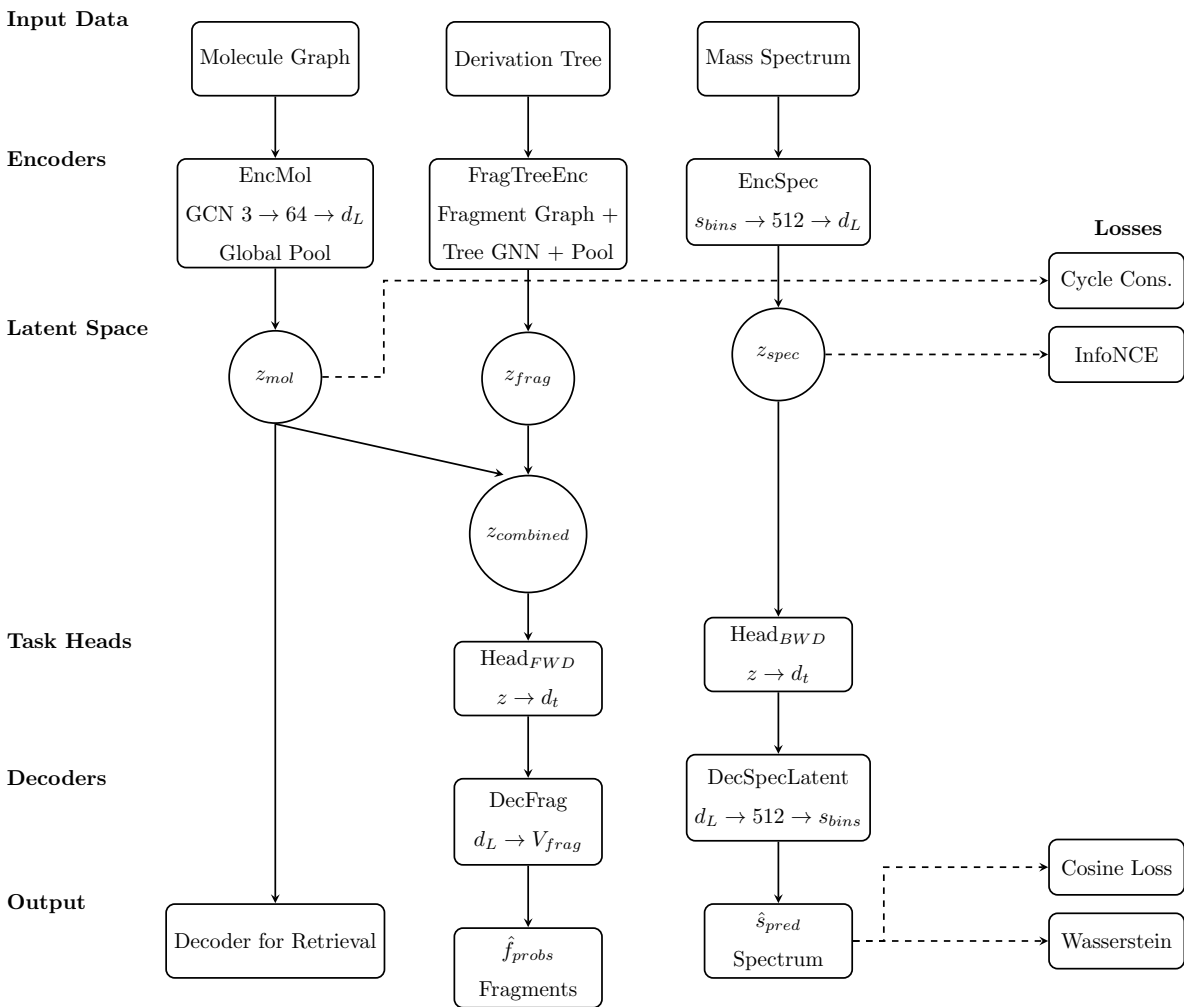


Figure 3.6: Diagram illustrating the overall architecture of the bidirectional translation framework, showing the flow from molecular graph encoding to spectral embedding and back.

3.4.2 Forward Model: Structure-to-Spectrum Prediction

The forward branch corresponds to the left-to-right pathway in Figure 3.6. Its purpose is to predict a mass spectrum from a molecular structure together with fragment-level information derived from precomputed derivation graphs. The molecular encoder, **EncMol**, is implemented as a two-layer graph convolutional network. It applies graph convolutions with dimensions $3 \rightarrow 64 \rightarrow d_L$ followed by global mean pooling, thereby mapping the input molecular graph to a latent vector z_{mol} . This matches the molecular branch depicted in the figure.

The fragment branch is conceptually represented by the *Derivation Graph* and *FragTreeEnc* boxes in Figure 3.6. In the implementation, this branch is realized by the wrapper module **FragSetEncoderWrapper**, which encodes derivation-graph structure explicitly. For each molecule, the dataset constructs fragment graphs and an explicit derivation graph whose nodes correspond to fragments and whose directed edges encode fragmentation relations. The fragment encoder first embeds the individual fragment graphs using the molecular graph encoder as a backbone. Fragment masses are appended as additional node features, and a second graph neural network is then applied over the derivation graph itself. A global pooling step yields a single latent fragment representation z_{frag} . Accordingly, this branch is implemented as a graph encoder over derivation graphs, not as a simple pooling operation over an unordered fragment bag.

The latent fusion step shown in Figure 3.6 as z_{combined} is implemented by a simple arithmetic average of the molecular and fragment latents,

$$z_{\text{combined}} = \frac{z_{\text{mol}} + z_{\text{frag}}}{2}.$$

This fused representation is then passed through the forward task head, **Head_{FWD}**, which projects the latent vector into a task-specific subspace. Finally, the shared decoder **DecSpecLatent** maps this representation to a predicted spectrum $\hat{s}_{\text{pred}}$. The forward branch is trained with the spectrum decoder as its decoder.

3.4.3 Backward Model: Spectrum-to-Structure Reconstruction

The backward branch corresponds to the spectrum-side pathway in Figure 3.6. Its function is to encode spectra into a latent representation that can be aligned with molecular embeddings and used for later retrieval. The spectrum encoder, **EncSpec**, receives a fixed-length vector of binned intensities. This encoder is implemented as a regularized stack of residual MLP blocks with normalization and dropout. This makes the spectral branch deeper and more expressive than the simplified notation in the figure might suggest.

Given an input spectrum, **EncSpec** produces a latent vector z_{spec} , which is then passed through the backward task head, **Head_{BWD}**. This head serves two purposes. First, it provides a task-specific projection for latent alignment with molecular embeddings during training. Second, it defines the representation that is used at test time for inverse retrieval. The projected spectrum embedding is compared to a latent index of molecular embeddings, allowing the model to retrieve the most similar candidate structures in embedding space.

It is worth emphasizing that the backward model is not a direct structure decoder in the generative sense. Instead, it is a spectrum-to-latent-to-candidate pipeline. The model first retrieves candidate molecules by cosine similarity in latent space and then optionally re-ranks them by passing each candidate through the forward branch and comparing the reconstructed spectrum with the query spectrum. Consequently, the "reconstruction" in this subsection should be interpreted as a process of structure retrieval guided by shared embeddings and forward-model consistency, rather than as the explicit generation of a graph.

3.4.4 Bidirectionality

The bidirectional nature of the framework arises from the fact that molecular graphs and spectra are not learned independently. As shown in Figure 3.6, both modalities are mapped into related latent spaces and are connected by shared components. The most important of these are the task heads and the spectrum decoder. The forward pathway employs a combination of molecular and fragment representations to predict spectra. In contrast, the backward pathway utilizes spectral embeddings to align against molecular embeddings and to facilitate retrieval. Because both directions are trained jointly within the same architecture, information learned in one direction constrains the other.

This bidirectionality is also strengthened by the retrieval strategy. The molecular encoder is used to build an index over known candidate structures, while the spectrum encoder provides the query representation. The result is that forward prediction and inverse retrieval are coupled tasks of the two views of the same embedding problem. In addition, the inverse direction can benefit from the forward model through reranking, since candidate molecules are rescored according to how well their forward-predicted spectra match the observed spectrum. In this sense, the model closes the loop between both directions even though it does not implement a fully generative cycle from spectrum directly back to a reconstructed molecular graph.

3.4.5 Loss shaping

The training objective is deliberately multi-component and is best understood as a form of loss shaping that stabilizes learning and injects chemically meaningful constraints. For the forward branch, the primary reconstruction objective combines cosine similarity with a one-dimensional Wasserstein distance between predicted and true spectra. The cosine term emphasizes agreement in spectral shape, while the Wasserstein term additionally penalizes positional mismatch along the ordered m/z axis. The combination is more informative than cosine similarity alone and aligns well with the loss panel outlined in Figure 3.6.

A second important component is the precursor-mass constraint. This forbidden-region penalty acts as a chemically motivated regularizer that penalizes predicted intensity above the precursor mass and suppresses implausible peaks. The forward phase also includes a cycle-style consistency term: a predicted spectrum is re-encoded by the spectrum encoder and encouraged to remain close to the molecular latent representation. In addition, a diversity loss discourages collapse toward overly similar spectral predictions within a batch.

For the backward branch, latent alignment is driven by an InfoNCE contrastive objective between spectrum and molecule embeddings. This enforces that matching molecule-spectrum pairs are close in latent space while non-matching pairs are separated. The backward phase further includes spectrum reconstruction from the spectral latent, the same precursor-mass penalty, additional cycle-consistency terms, and an explicit retrieval-ranking loss that rewards within-batch discrimination of the correct molecular candidate. Finally, selected losses are uncertainty-weighted through learned log-variance parameters in the task-head module.

Taken together, these terms show that the architecture in Figure 3.6 should not be interpreted as a simple encoder-decoder stack. Rather, it is a coupled bidirectional system in which reconstruction, alignment, ranking, and chemical plausibility constraints are optimized simultaneously. This loss shaping is essential for turning the shared latent space into a representation that is useful both for forward spectrum prediction and for backward structure retrieval.

Chapter 4

Results and Discussion

4.1 Computation Times of Fragmentation Expansion

Computation times present a significant bottleneck for this approach. The `mod` framework, while powerful in generating comprehensive reaction networks, prioritizes flexibility and completeness over computational efficiency. Combined with the combinatorial explosion of possible fragments and parent molecules, runtime becomes prohibitive as the number of fragmentation rules increases. Since only a subset of known rules were applied in this study, computation times per molecule will rise substantially as additional rules are incorporated, making further optimization necessary to ensure the feasibility of this approach at scale.

4.1.1 Visualization of Computation Times

A regression analysis was performed to model the relationship between molecular complexity and computation time. The analysis assumes:

- Homoscedasticity of residuals
- Independence of residual observations
- Normality of residual distribution

In Figure 4.1, the regression results are visualized, showing a positive correlation between molecular complexity and computation time, but also show a large variance in computation times for molecules of similar complexity. Note that the threshold of 40 seconds was chosen arbitrarily to exclude trivial expansions.

So while there is a general trend of increasing computation time with molecular complexity, the large variance suggests that other factors, such as specific structural features or the particular fragmentation pathways available, also play a significant role in determining computation time.

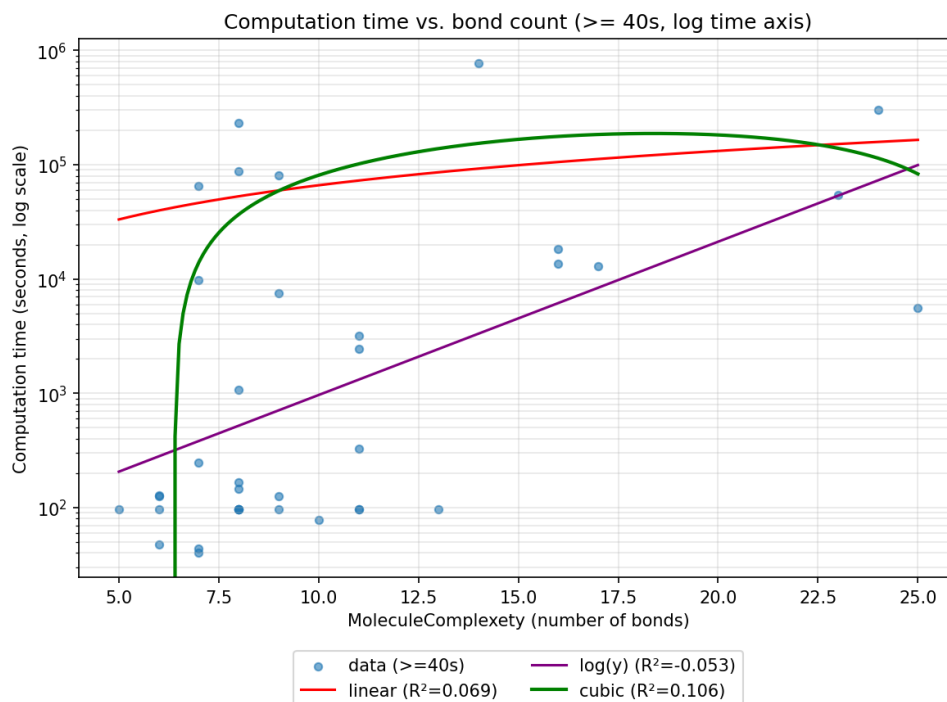


Figure 4.1: Regression of Computation time, excluding molecules that had trivial expansions (time < 40s). The plot shows a positive correlation between molecular complexity and computation time, but also a large variance in computation times for molecules of similar complexity.

This highlights the need for further optimization.

4.1.2 Data Splits

The dataset was partitioned in two stages into training, validation, and test subsets. In the first stage, the full set of molecule-spectrum pairs was divided into an 80% training portion and a 20% test portion. In the second stage, the initial training portion was further split into 70% training and 30% validation. This results in a final distribution of 56% training data, 24% validation data, and 20% test data relative to the original dataset.

These ratios were chosen as a pragmatic trade-off between model development and robust evaluation. The 80/20 split keeps a sufficiently large hold-out test set for a meaningful final performance estimate, while still reserving most samples for development. Splitting the development portion into 70/30 then provides enough training data for parameter learning and enough validation data for stable model selection and hyperparameter tuning.

This splitting strategy serves two purposes. First, it preserves a dedicated hold-out test set that is never used during model development and is therefore reserved for the final evaluation. Second, it provides a separate validation set for model selection, hyperparameter tuning, and intermediate performance monitoring. In this way, the validation set supports iterative model

development, while the test set remains an unbiased reference for the final assessment.

All splits are generated with a fixed random seed (`random_state=42`) to ensure reproducibility. At the same time, the implementation uses `shuffle=False`, so the split is deterministic and preserves the original order of the dataset rather than randomly permuting the samples beforehand. This design choice has to be interpreted carefully. On the one hand, it guarantees that repeated runs use exactly the same partitioning. On the other hand, if the original ordering of molecules is not random, the resulting subsets may differ systematically in composition. The reported results should therefore be understood with this deterministic split procedure in mind.

An additional technical constraint concerns the fragment representation. The fragment vocabulary is constructed only from the training set and then reused unchanged for the validation and test sets. This ensures that the dimensionality of fragment-based model components remains fixed across all stages of training and evaluation. It also prevents information leakage from validation or test data into the feature space used during training. In practical terms, this means that fragment identifiers available during evaluation are restricted to those observed in the training data, which is consistent with a realistic supervised learning setting.

Overall, the chosen split strategy provides a clear separation between model fitting, model selection, and final evaluation, while maintaining a consistent fragment representation across all subsets. At the same time, the use of a deterministic non-shuffled split should be considered when interpreting the generality of the reported results.

4.2 Evaluation of Fragment Prediction

The evaluation reveals a mixed but informative picture of the proposed framework. On the one hand, the model learns a usable mapping between molecular structures and spectra in both directions. On the other hand, the fragment-aware components do not currently provide a clear empirical advantage over simpler ablated variants. The strongest positive result instead arises from the reranking stage, which substantially improves retrieval quality. To isolate the contribution of individual components, this section includes an *ablation study* comparing the full model against no-fragments, no-derivation-tree, and no-reranking variants. The paired randomization tests underlying this comparison did not yield statistically significant differences for any of the pairwise contrasts, with all p-values exceeding 0.05.

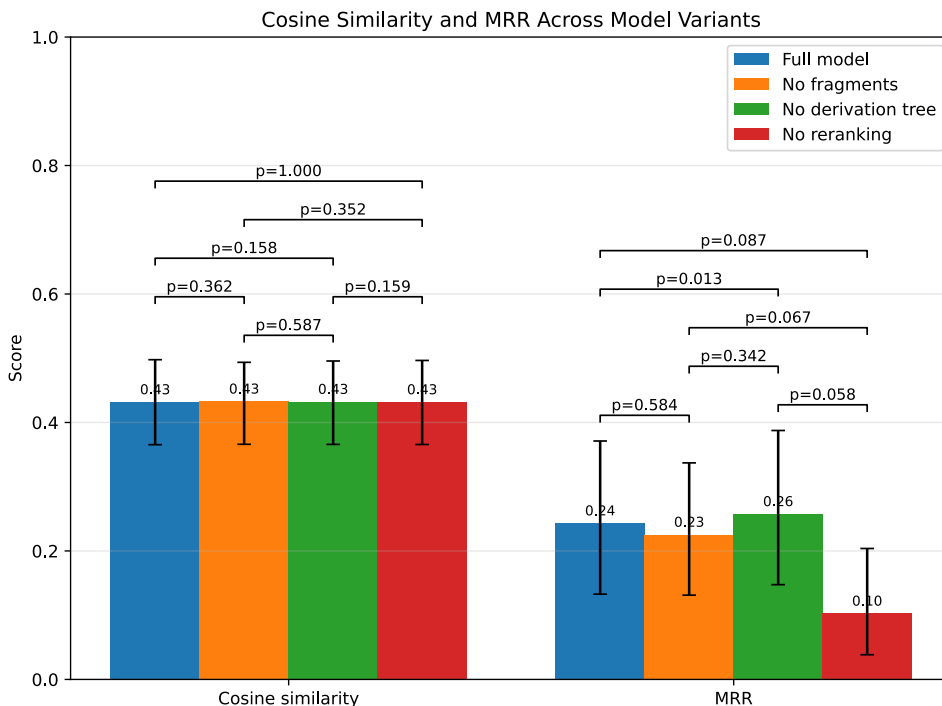


Figure 4.2: Ablation of three pipeline components (fragments, derivation tree, and reranker) against the full model. Cosine similarity measures how similar the query is to the correct target, averaged over all queries; MRR is the mean reciprocal rank of the ground-truth target. Error bars show 95% bootstrap confidence intervals (5000 query resamples). Brackets indicate pairwise p-values from paired randomization tests computed separately for each metric. Cosine similarity remains nearly constant across variants (≈ 0.43), while MRR decreases from 0.24 to 0.10 when reranking is removed. Since all confidence intervals overlap and all pairwise p-values exceed 0.05, these differences should be interpreted as suggestive rather than statistically established.

In this ablation study, with respect to the forward task, all evaluated variants achieve very similar cosine similarity values. The full model reaches a mean cosine similarity of 0.4316, while the no-fragments and no-derivation-tree ablations achieve 0.4327 and 0.4322, respectively. Which is visible in Figure 4.2. These differences are extremely small and indicate that the forward reconstruction quality is largely insensitive to the inclusion or removal of the current fragment-aware branch. In practical terms, the molecular encoder appears to capture most of the predictive signal required for the structure-to-spectrum task, while the fragment representation does not yet provide an additional measurable benefit.

The retrieval results are more informative. As shown in Figure 4.3, for the full model, Recall@1 is 0.091, Recall@5 is 0.318, Recall@10 is 0.500, and Recall@20 is 0.864, with a mean reciprocal rank (MRR) of 0.243. This shows that the correct molecule is frequently brought into the top portion of the ranked candidate list, although top-1 identification remains limited.

The model therefore behaves more like a candidate narrowing system than a reliable single-shot identifier, which is still useful in practical structure elucidation workflows.

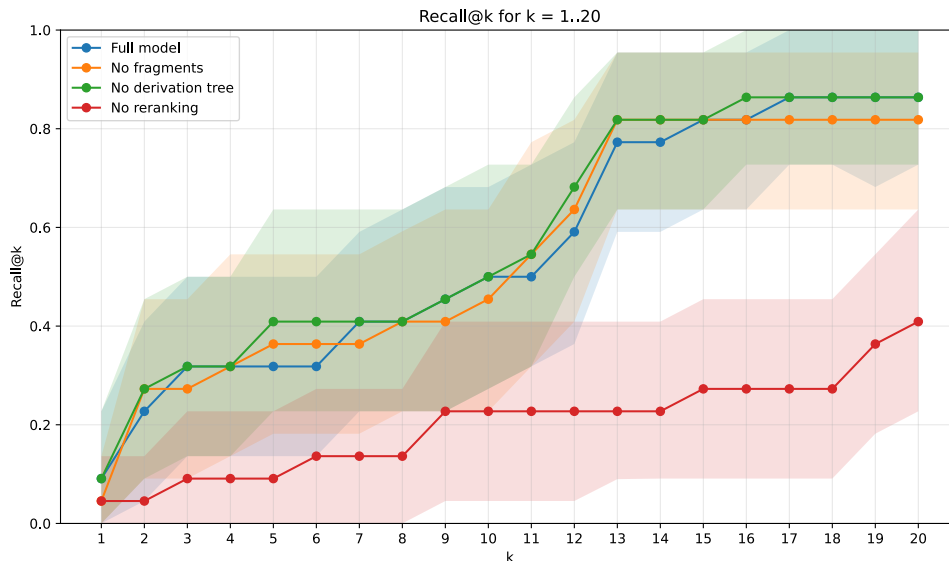


Figure 4.3: Recall@ k for $k = 1, \dots, 20$ across the full retrieval model and three ablated variants. Shaded bands show 95% bootstrap confidence intervals over queries. The full model and the **no fragments** / **no derivation tree** variants cluster together across the full k range, while the **no reranking** variant trails the others at every k , with the largest gap at small k and gradual closure toward $k = 20$.

The clearest effect in the evaluation is observed for the reranking stage. Removing reranking leaves the forward cosine similarity unchanged, as expected, but substantially reduces retrieval quality. Recall@5 drops from 0.318 to 0.091, Recall@10 from 0.500 to 0.227, Recall@20 from 0.864 to 0.409, and MRR from 0.243 to 0.103. This demonstrates that much of the retrieval benefit of the framework arises from combining latent-space retrieval with forward-model-based rescoring. In this sense, the present implementation benefits more from the interaction between the forward and backward branches than from the fragment-aware representation itself.

By contrast, the fragment-related ablations do not support a strong performance claim. The no-fragments variant performs slightly better than the full model on cosine similarity and Recall@5, but slightly worse on Recall@10, Recall@20, and MRR. Similarly, the no-derivation-tree variant is highly competitive and even slightly exceeds the full model on Recall@5 and MRR. These results suggest that the fragment branch, in its current form, is not yet contributing a robust and clearly positive signal. At present, it appears to be largely redundant with the molecular representation, and may in some cases introduce noise rather than useful structure.

This conclusion should, however, be interpreted cautiously. The test set contains only 22

queries, so recall values change in increments of approximately 0.045. Consequently, differences between variants often reflect only one or two queries. Query-level paired analyses confirm that the differences between the full model and the no-fragments ablation are not statistically convincing for the main metrics, whereas the benefit of reranking is much more substantial. The present evaluation therefore provides stronger support for the usefulness of reranking than for the effectiveness of the fragment-aware branch.

Overall, the results show that the proposed bidirectional framework is functional and that reranking substantially improves retrieval performance. However, they also indicate that the current fragment integration does not yet realize its intended benefit. The main contribution of the present work is therefore not the demonstration of a clear performance gain from fragment augmentation, but rather the establishment of a fragment-augmented bidirectional framework whose strengths, weaknesses, and failure modes can be analyzed systematically. This in turn provides a concrete basis for future work aimed at improving the informativeness and integration of fragment-level intermediate representations.

4.3 Challenges

The thesis is shaped by several interconnected challenges.

First, fragmentation must be represented in a way that is simultaneously chemically meaningful and useful for machine learning.

Second, explicit fragment generation leads to a *combinatorial explosion* of candidate structures and limits the practical scale of the approach.

Third, the absence of unique *ground-truth fragmentation trees* makes fragment-level supervision inherently *ambiguous*.

Fourth, bidirectional learning introduces *competing objectives*, since forward spectrum reconstruction and backward structure retrieval do not necessarily favor the same latent representation.

Fifth, evaluation is intrinsically multi-dimensional, requiring predictive, retrieval-based, and interpretability-oriented criteria.

Finally and **sixth**, the empirical contribution of fragment-aware intermediate representations remains difficult to establish conclusively, especially under *small-sample* conditions and in the presence of strong baseline signals from the molecular encoder alone.

Chapter 5

Future Work

A central limitation of the present framework is that it does not learn fragmentation generation itself. In its current form, the model operates on *precomputed* derivation graphs: candidate fragments and their parent-child relations are generated upstream by the rule-based graph-transformation pipeline and are then encoded by the neural model. In other words, the learning problem begins only *after* a fragmentation space has already been constructed. The neural architecture therefore learns how to embed, combine, align, and score molecular, fragment, and spectral information, but it does not yet learn which fragmentation events should be proposed in the first place. This design is appropriate for a first proof of concept, since it makes the fragment-level representation explicit and keeps the learning problem tractable. At the same time, it also marks the clearest boundary of the current approach and thus the most important direction for future work.

5.1 From Fragment Encoding to Fragment Generation

The present implementation is best understood as a *fragment-aware encoder* rather than a full fragmentation simulator. Its fragment branch consumes fragment graphs together with their derivation structure, for example through the derivation-graph backend implemented in `build_derivation_graph_from_collection` and `FragSetEncoderWrapper`. These components construct and encode a graph whose nodes are fragment graphs and whose directed edges represent parent-child relations extracted from the precomputed derivation graph. This means that the model receives a structured fragmentation hypothesis as input instead of having to infer it autonomously. A natural next step would therefore be to move from *encoding* fragmentation structures to *generating* them. Conceptually, this would shift the current system from a hybrid model with symbolic preprocessing and neural downstream learning toward a more integrated

neuro-symbolic architecture. Rather than assuming that the relevant fragment space has already been enumerated, the model could be trained to predict which bonds or substructures are most likely to fragment, in which order, and under which contextual conditions. Such a development would make the system considerably closer to a true mechanistic simulator of fragmentation pathways.

5.2 Toward a Generate–Score Architecture

An important source of inspiration for this transition is ICEBERG [10], which explicitly separates the task into two components: a *Generate* module that predicts likely fragmentation events and a *Score* module that assigns intensities to the resulting fragments. The key contribution of that formulation is that it does not treat the fragment graph as fixed input, but as a latent object that must itself be predicted. ICEBERG is described as a two-part model in which one component generates molecular substructure candidates and a second component predicts intensities and hydrogen rearrangements for those generated fragments. It further trains the generative stage to reconstruct an estimated fragmentation DAG and uses a second module to score the resulting candidate fragments. Adapting this idea to the present framework would be highly promising. The current model already contains the main ingredients needed for the *Score* side of such a system: it embeds molecular graphs, fragment-level structures, and spectra into a shared latent space; it predicts spectra from latent representations; and it uses bidirectional training signals such as contrastive alignment and reconstruction losses. What is missing is an upstream neural module that decides which fragments should enter the fragment branch at all. In future work, the symbolic derivation graphs produced by the rule-based pipeline could serve as supervision for such a generator. Initially, this could be implemented as a learned pruning mechanism over the derivation graph, followed by a more ambitious autoregressive model that predicts fragmentation events directly.

5.3 Advantages of Learning Fragment Generation

Learning fragmentation generation would offer several advantages over the present setup. First, it would reduce the model’s dependence on exhaustive or semi-exhaustive preprocessing. At the moment, the symbolic pipeline must already generate a fragmentation space before learning begins. This introduces computational cost and also constrains the neural model to the candidate set produced upstream. A learned generator could instead prioritize only the most informative or probable fragmentation events, thereby reducing the combinatorial burden and potentially

improving scalability. Second, a generative fragmentation stage would make the architecture more *end-to-end*. Currently, the spectrum predictor is conditioned on a fragment representation that is treated as given. In a more integrated design, fragment proposals, fragment embeddings, and spectrum scoring could be optimized jointly. This would make it possible for the model to learn not only how to use fragments, but also which fragment structures are most helpful for downstream spectrum reconstruction and inverse retrieval. Third, explicit fragmentation generation may improve interpretability. The current framework is interpretable insofar as it operates on chemically meaningful fragment structures. However, because those structures are externally generated, the model cannot explain why one fragmentation path was preferred over another. A learned generative stage could provide atom-wise or bond-wise probabilities over fragmentation events and thereby yield a more direct account of the model’s mechanistic assumptions.

5.4 Scientific and Technical Challenges

This transition is non-trivial and raises several challenges. The first is the need for supervision. Spectral libraries generally provide molecule–spectrum pairs, but not ground-truth fragmentation trees. ICEBERG addresses this by using a heuristic, combinatorially generated fragmentation DAG as a training target. A similar strategy would likely be required here: the current derivation graphs generated by the rule-based MØD pipeline could serve as proxy labels for a learned generator. However, such labels are not guaranteed to be chemically unique or physically exact, and this ambiguity would need to be addressed carefully during training and evaluation. A second challenge concerns error propagation. In the current architecture, the fragment encoder assumes that the fragment graph is already available. Once generation becomes learned, mistakes made by the generative stage may propagate into the spectral scoring stage. This suggests that future work should not simply replace symbolic enumeration with a neural generator, but should consider staged training, beam search, uncertainty estimation, or latent-variable formulations that can maintain several plausible fragmentation hypotheses. A third challenge is chemical realism. A learned generator should ideally distinguish between fragments that are structurally possible and fragments that are spectrometrically plausible, including effects such as charge retention, neutral loss, and hydrogen rearrangements. This again argues in favor of a modular Generate–Score design rather than a single black-box network.

5.5 A Plausible Development Path

A realistic path forward would proceed in stages. The first stage would keep the current derivation graphs but train a neural module to rank or prune candidate fragmentation steps. The second stage would learn to expand fragmentation trees autoregressively, conditioned on the precursor graph and the partial fragmentation history. The third stage would couple this generator tightly to the existing scoring model so that fragment generation, spectral prediction, and inverse retrieval are optimized jointly. In this way, the current system can be seen not as an alternative to Generate-Score approaches, but as an intermediate step toward them. In summary, the most important future extension of the present work is to shift the model from *fragment-structure encoding* to *fragmentation-process learning*. The current architecture already demonstrates that fragment-level intermediate representations can improve the connection between molecules and spectra. The next major advance will be to let the model learn which fragmentation structures to construct, rather than merely how to encode those supplied by an external pipeline. This would bring the framework closer to a fully trainable, mechanistically informed, and end-to-end bidirectional model for structure-spectrum translation.

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Appendix A

Source Code Appendix

A.1 Github Repository

The code for this thesis is available at <https://github.com/phireiser/mass-spec>